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ODISHA BHUBANESWAR INDIA

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C. V. Raman Global University
ODISHA BHUBANESWAR INDIA

Department of Chemistry

BIOLOGY AND ENVIRONMENTAL SCIENCE (CY202)

UNIT-V : Water Pollution Abatement and Solid Waste Management

Dr. Gulmi Chakraborty
Assistant Professor

Outline

01 Structure of Ecosystem

02 Air and Water Pollution: sources and treatment

03 Solid waste, Noise pollution and other hazards: Sources and mitigation techniques

Water Quality Parameters

- Water is the second most important need for life to exist after air.
- The most popular definition of water quality is “it is the physical, chemical, and biological characteristics of water”.
- Based on its source, water can be divided into ground water and surface water.
- Both types of water can be exposed to contamination risks from agricultural, industrial, and domestic activities, which may include many types of pollutants such as heavy metals, pesticides, fertilizers, hazardous chemicals, and oils.
- Parameters that are frequently sampled or monitored for water quality include temperature, dissolved oxygen, pH, conductivity, BOD, TOC and turbidity.

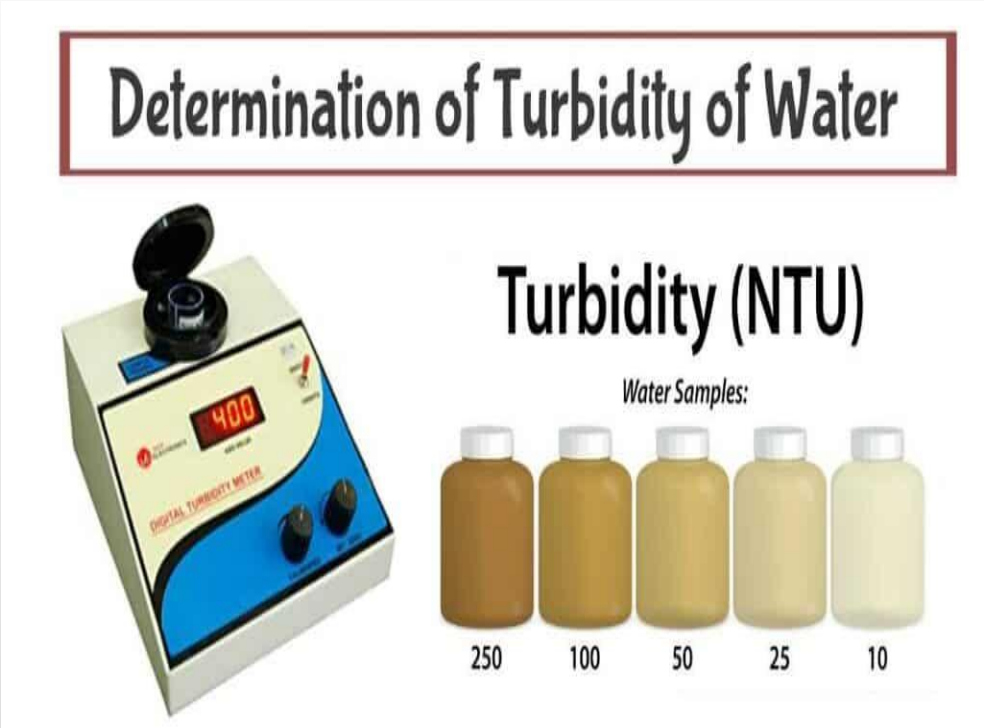
Physical parameters of water quality

There are three types of water quality parameters physical, chemical, and biological

Physical parameters of water quality

Turbidity

- Turbidity is the cloudiness of water.
- It is a measure of the ability of light to pass through water.
- It is caused by suspended material such as clay, silt, organic material, plankton, and other particulate materials in water.
- Turbidity in drinking water is esthetically unacceptable, which makes the water look unappetizing.



https://civiconcepts.com/blog/determination-of-turbidity-of-water-experiment

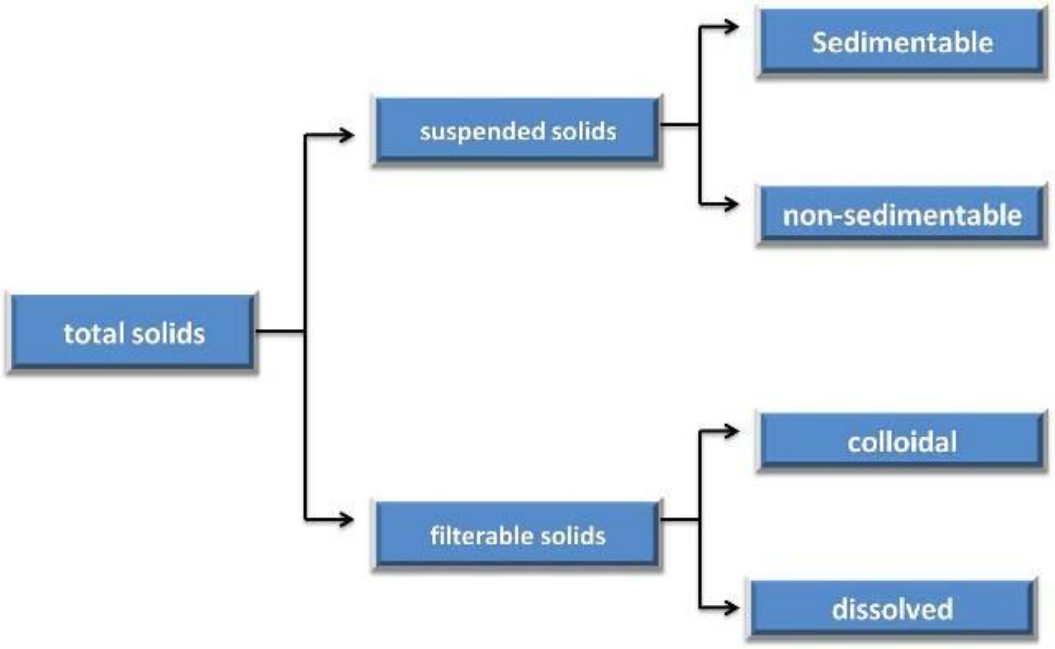
Physical parameters of water quality

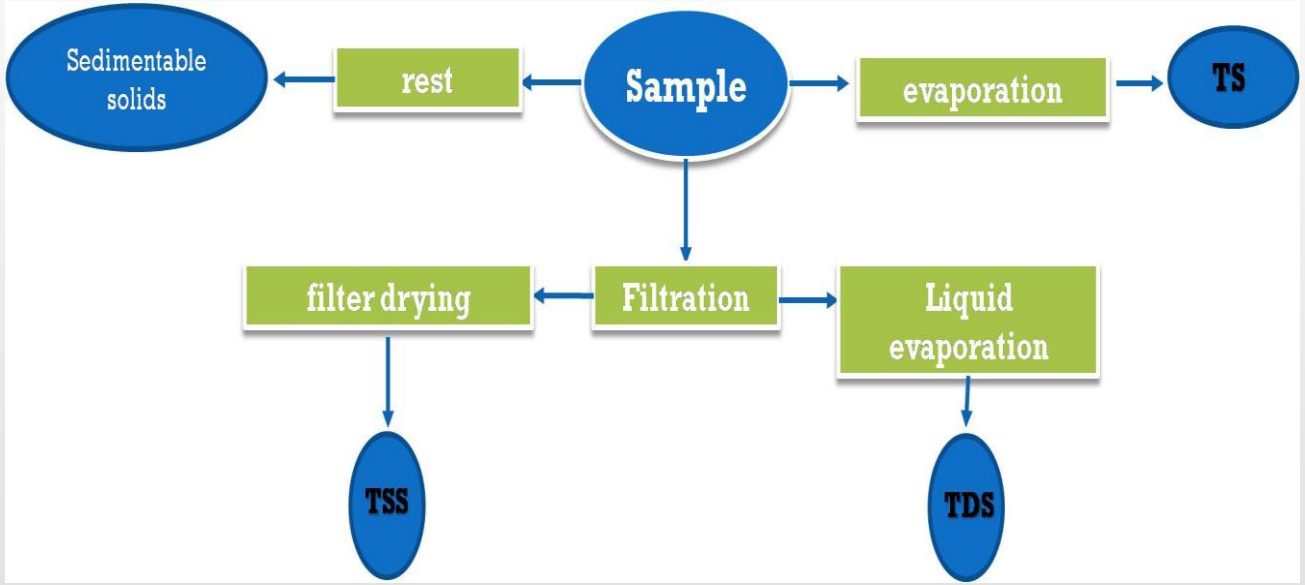
- The impact of turbidity can be summarized in the following points:
- It can increase the cost of water treatment for various uses.
- The particulates can provide hiding places for harmful microorganisms and thereby shield them from the disinfection process.
- Suspended materials can clog or damage fish gills, decreasing its resistance to diseases, reducing its growth rates, affecting egg and larval maturing, and affecting the efficiency of fish catching method.
- Suspended particles provide adsorption media for heavy metals such as mercury, chromium, lead, cadmium, and many hazardous organic pollutants such as polychlorinated biphenyls (PCBs), polycyclic aromatic hydrocarbons (PAHs), and many pesticides.

Physical parameters of water quality

- The amount of available food is reduced because higher turbidity raises water temperatures in light of the fact that suspended particles absorb more sun heat.
- Consequently, the concentration of the dissolved oxygen (DO) can be decreased since warm water carries less dissolved oxygen than cold water.
- Turbidity is measured by an instrument called nephelometric turbidimeter, which expresses turbidity in terms of NTU or TU. A TU is equivalent to 1 mg/L of silica in suspension.
- Turbidity more than 5 NTU can be visible to the average person while turbidity in muddy water, it exceeds 100 NTU.
- Groundwater normally has very low turbidity because of the natural filtration that occurs as the water penetrates through the soil

Physical parameters of water quality





https://hive.blog/hive-196387/@yusvelasquez/determination-of-total-solids-in-water-samples

Physical parameters of water quality

https://www.youtube.com/watch?v=8Yf9yhtH28g

Physical parameters of water quality

Solids

Solids occur in water either in solution or in suspension. These two types of solids.

By definition, the suspended solids are retained on the top of the filter and the dissolved solids pass through the filter with the water.

If the filtered portion of the water sample is placed in a small dish and then evaporated, the solids as a residue. This material is usually called total dissolved solids or TDS.

Total solid(TS)=Total dissolved solid(TDS)+Total suspended solid (TSS)



https://hive.blog/hive-196387/@yusvelasquez/determination-of-total-solids-in-water-samples

Physical parameters of water quality

Solids

Water can be classified by the amount of TDS per liter as follows:

freshwater: <1500 mg/L TDS;

brackish water: 1500–5000 mg/L TDS;

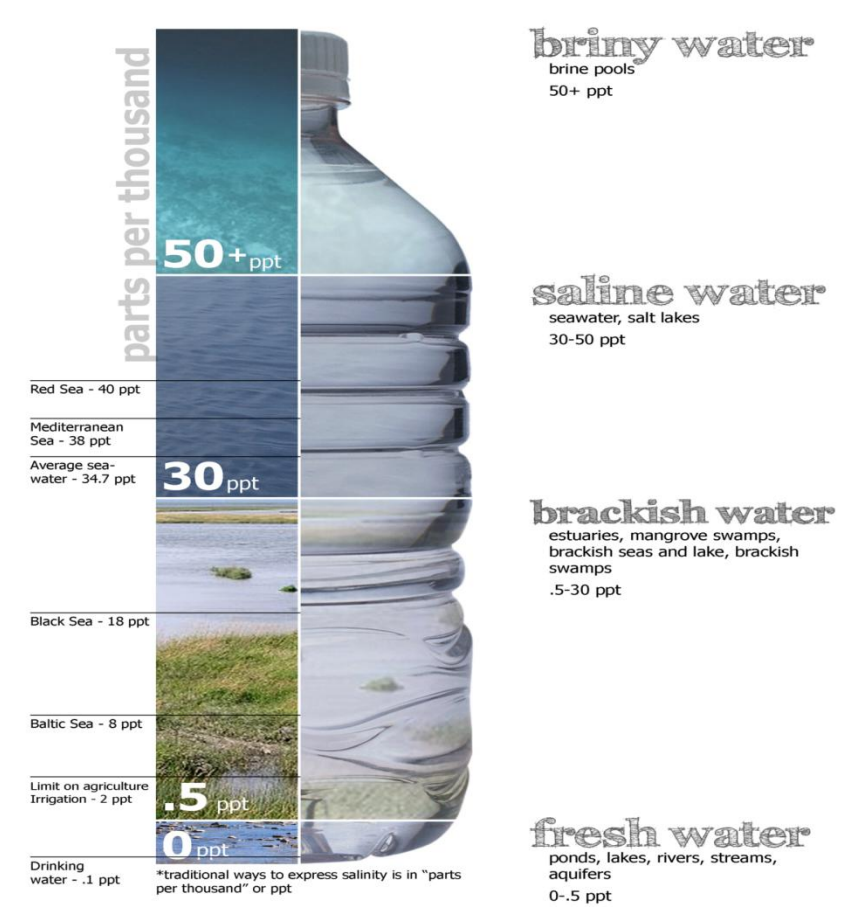
saline water: >5000 mg/L TDS.

The residue of TSS and TDS after heating to dryness for a defined period of time and at a specific temperature is defined as fixed solids. Volatile solids are those solids lost on ignition (heating to 550°C) .

Physical parameters of water quality

- **Brackish water**, sometimes termed **brack water**, is water occurring in a natural environment that has more salinity than freshwater, but not as much as seawater.
- It may result from mixing seawater (salt water) and fresh water together, as in estuaries, or it may occur in brackish fossil aquifers.
- Because brackish water is hostile to the growth of most terrestrial plant species, without appropriate management it is damaging to the environment.
- Technically, brackish water contains between 0.5 and 30 grams of salt per litre—more often expressed as 0.5 to 30 parts per thousand (‰), which is a specific gravity of between 1.0004 and 1.0226.
- Thus, *brackish* covers a range of salinity regimes and is not considered a precisely defined condition.
- It is characteristic of many brackish surface waters that their salinity can vary considerably over space or time.

Physical parameters of water quality



https://en.wikipedia.org/wiki/Brackish_water



https://shop16430.heretal.com/category?name=salt%20water%20meets%20fresh%20water%20location

Physical parameters of water quality

Electrical conductivity (EC)

The electrical conductivity (EC) of water is a measure of the ability of a solution to carry or conduct an electrical current. Since the electrical current is carried by ions in solution, the conductivity increases as the concentration of ions increases. Therefore, it is one of the main parameters used to determine the suitability of water for irrigation and firefighting.

Units of its measurement are as follows:

U.S. units = micromhos/cm

S.I. units = milliSiemens/m (mS/m) or dS/m (deciSiemens/m)

where (mS/m) = 10 umho/cm (1000 µS/cm = 1 dS/m).

Physical parameters of water quality

The electrical conductivity can be used to estimate the TDS value of water as follows:

TDS (mg/L) = k_e × EC (µS/cm) where k_e is a constant of proportionality

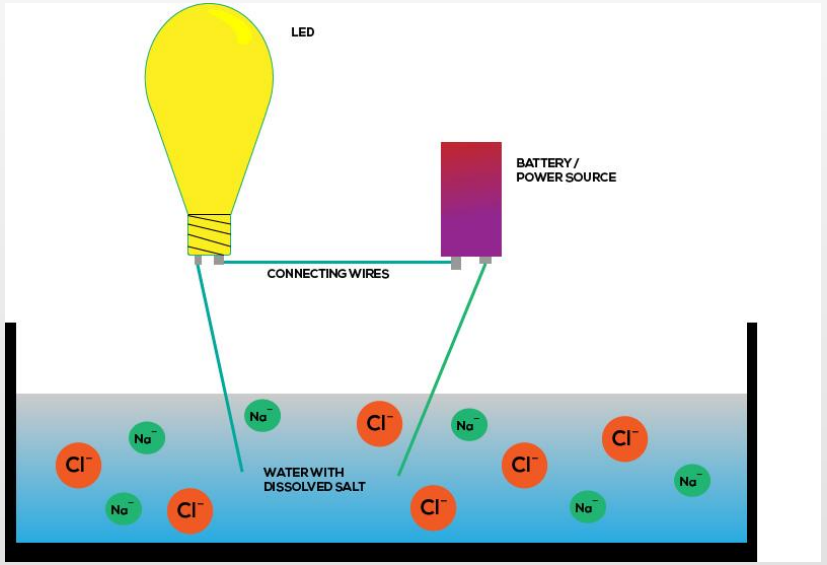
TDS can be used to estimate the ionic strength of water in the applications of groundwater recharging by treated wastewater. The normal method of measurement is electrometric method.

Pure water is not a good conductor of electricity. Typical conductivity of water is as follows:

Ultra-pure water: 5.5×10^{-6} S/m;

Drinking water: 0.005–0.05 S/m;

Seawater: 5 S/m.

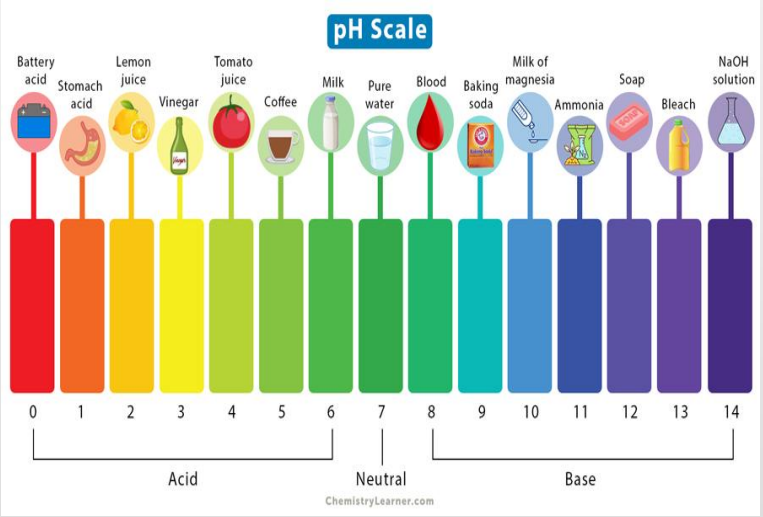


https://www.greenspan.com.au/2021/05/06/electrical-conductivity/

Chemical parameters of water quality

pH

- It is defined as the negative logarithm of the hydrogen ion concentration.
- It is a dimensionless number indicating the strength of an acidic or a basic solution. Actually, pH of water is a measure of how acidic/basic water is.
- Acidic water contains extra hydrogen ions (H^+) and basic water contains extra hydroxyl (OH^-) ions.
- Pure water is neutral, with a pH close to 7.0 at 25°C.
- Normal rainfall has a pH of approximately 5.6 (slightly acidic) owing to atmospheric carbon dioxide gas.
- Safe ranges of pH for drinking water are from 6.5 to 8.5 for domestic use and living organisms need.



https://www.chemistrylearner.com/ph-scale.html

Parameters of water quality

- The effects of pH on animals and plants can be summarized as follows:
- Water with very low or high pH is fatal.
- A pH below 4 or above 10 will kill most fish, and very few animals can endure water with a pH below 3 or above 11.
- The low pH may be due to low pH levels induced by acid rain.
- Heavy metals such as cadmium, lead, and chromium dissolve more easily in highly acidic water (lower pH).
- This is important because many heavy metals become much more toxic when dissolved in water.

Dissolved Oxygen

- Dissolved oxygen (DO) is considered to be one of the most important parameters of water quality in streams, rivers, and lakes. It is a key test of water pollution . The higher the concentration of dissolved oxygen, the better the water quality.
- Oxygen is slightly soluble in water and very sensitive to temperature. For example, the saturation concentration at 20°C is about 9 mg/L and at 0°C is 14.6 mg/L.
- The actual amount of dissolved oxygen varies depending on pressure, temperature, and salinity of the water. Dissolved oxygen has no direct effect on public health, but drinking water with very little or no oxygen tastes unpalatable to some people.
- There are three main methods used for measuring dissolved oxygen concentrations: the colorimetric method—quick and inexpensive, the Winkler titration method—traditional method, and the electrometric method.

Biochemical oxygen demand (BOD)

➢Bacteria and other microorganisms use organic substances for food. As they metabolize organic material, they consume oxygen. The organics are broken down into simpler compounds, such as CO_2 and H_2O, and the microbes use the energy released for growth and reproduction.

➢When this process occurs in water, the oxygen consumed is the DO in the water. If oxygen is not continuously replaced by natural or artificial means in the water, the DO concentration will reduce as the microbes decompose the organic materials. This need for oxygen is called the biochemical oxygen demand (BOD).

➢The more organic material there is in the water, the higher the BOD used by the microbes will be.

BOD Level (in ppm)	Water Quality
1-2	**Very Good** There will not be much organic waste present in the water supply.
3-5	**Fair: Moderately Clean**
6-9	**Poor: Somewhat Polluted** Usually indicates organic matter is present and bacteria are decomposing this waste."
100 or greater	**Very Poor: Very Polluted** ontains organic waste.

Biochemical oxygen demand (BOD)

- BOD is used as a measure of the power of sewage; strong sewage has a high BOD and weak sewage has low BOD.
- The complete decomposition of organic material by microorganisms takes time, usually 20 d or more under ordinary circumstances.
- The quantity of oxygen used in a specified volume of water to fully decompose or stabilize all biodegradable organic substances is called the ultimate BOD or BOD_L.
- BOD is a function of time. At time = 0, no oxygen will have been consumed and the BOD = 0. As each day goes by, oxygen is used by the microbes and the BOD increases. Ultimately, the BOD_L is reached and the organic materials are completely decomposed.

Biochemical oxygen demand (BOD)

Q.1 The results from a BOD test, diluted by 100, are given in columns 1 and 2 of the Table-1. Calculate BOD_5.

Ans: $BOD_5 = p (DO_I - DO_f)$, p = dilution of the sample in the BOD bottle (e.g. 100)

DO_I and DO_f initial and final DO concentration

Time (days)	Diluted sample DO (mg/L)	BOD (calculated) (mg/L)
0	7.95	------
1	3.75	420
3	2.75	520
5	1.80	615

$BOD_5 = 100 (7.95-1.8) = 615$ mg/L

Biochemical oxygen demand corrected (BOD)

Q.1 The results from a BOD test, diluted by 100, are given in columns 1 and 2 of the Table-1. Calculate BOD_5.

Ans: $BOD_5 = p\,(DO_I - DO_f)$, p = dilution of the sample in the BOD bottle (e.g. 100)

DO_I and DO_f initial and final DO concentration

Time (days)	Diluted sample DO (mg/L)	Blank seed sample DO (mg/L)	BOD (calculated) (mg/L)	BOD corrected (mg/L)
0	7.95		------	
1	3.75		420	
3	2.75		520	
5	1.80		615	

F = (% seed in DO_1/ % seed in B_1) =

In addition to the BOD of waste water the seeded dilution water has also some small BOD. Hence, corrected BOD becomes:

$BOD = [p(DO_I - DO_f) - (B_I - B_f)f]$, B, B = initial and final DO concentrations of the seeded diluted water (blanks)

f = ratio of seed DO in sample to seed in blanks = (% seed in DO_1/ % seed in B_1)

Chemical Oxygen Demand (COD)

- The chemical oxygen demand (COD) is a parameter that measures all organics: the biodegradable and the non-biodegradable substances.
- It is a chemical test using strong oxidizing chemicals (potassium dichromate), sulfuric acid, and heat, and the result can be available in just 2 h.
- COD values are always higher than BOD values for the same sample.

Water Pollution

- **water pollution** corresponds to the release of substances into subsurface groundwater or into lakes, streams, rivers, estuaries, and oceans to the point where the substances interfere with beneficial use of the water or with the natural functioning of ecosystems.
- It also refers to the release of substances, such as chemicals, trash, or microorganisms.
- water pollution may also include the release of energy, in the form of radioactivity or heat, into bodies of water.



https://trityenviro.com/bd/8-major-causes-of-water-pollution-understanding-the-sources-and-impacts-in-india/

Sources of Water Pollution

- Water pollutants come from either point sources or dispersed sources. A point source is a pipe or channel, such as those used for discharge from an industrial facility or a city sewage system.
- A dispersed (or nonpoint) source is a very broad unconfined area from which a variety of pollutants enter the water body, such as the runoff from an agricultural area.
- Point sources of water pollution are easier to control than dispersed sources, because the contaminated water has been collected and conveyed to one single point where it can be treated.
- Pollution from dispersed sources is difficult to control, and, despite much progress in the building of modern sewage-treatment plants, dispersed sources continue to cause a large fraction of water pollution problems.



https://www.nrdc.org/stories/water-pollution-everything-you-need-know



https://www.britannica.com/topic/dispersed-source-pollutant

Types of water pollution

Domestic sewage

➢Domestic sewage is the primary source of pathogens (disease-causing microorganisms) and putrescible organic substances. Because pathogens are excreted in feces, all sewage from cities and towns is likely to contain pathogens of some type, potentially presenting a direct threat to public health.

➢Domestic sewage is also a major source of plant nutrients, mainly nitrates and phosphates. Excess nitrates and phosphates in water promote the growth of algae, sometimes causing unusually dense and rapid growths known as algal blooms.



https://www.wavetribe.com/blogs/eco/colors-of-the-ocean-why-are-some-oceans-blue-and-others-green

Types of water pollution

- When the algae die, oxygen dissolved in the water declines because microorganisms use oxygen to digest algae during the process of decomposition. Anaerobic organisms (organisms that do not require oxygen to live) then metabolize the organic wastes, releasing gases such as methane and hydrogen sulfide, which are harmful to the aerobic (oxygen-requiring) forms of life.
- The process by which a lake changes from a clean, clear condition—with a relatively low concentration of dissolved nutrients and a balanced aquatic community—to a nutrient-rich, algae-filled state and thence to an oxygen-deficient, waste-filled condition is called eutrophication.
- Eutrophication is a naturally occurring, slow, and inevitable process.
- However, when it is accelerated by human activity and water pollution (a phenomenon called cultural eutrophication), it can lead to the premature aging and death of a body of water.

Eutrophication



https://whalecoastconservation.org.za/eutrophication



https://energyeducation.ca/encyclopedia/Eutrophication

Solid waste

➢The improper disposal of solid waste is a major source of water pollution. Solid waste includes garbage, rubbish, electronic waste, trash, and construction and demolition waste, all of which are generated by individual, residential, commercial, institutional, and industrial activities.

➢Land pollution can also become water pollution if the trash or other debris is carried by animals, wind, or rainfall to bodies of water. Significant amounts of solid waste pollution in inland bodies of water can also eventually make their way to the ocean. Solid waste pollution is unsightly and damaging to the health of aquatic ecosystems and can harm wildlife directly. Many solid wastes, such as plastics and electronic waste, break down and leach harmful chemicals into the water, making them a source of toxic or hazardous waste.



https://eartheclipse.com/environment/pollution/amazing-ways-to-prevent-water-pollution.html

Toxic waste

- Waste is considered toxic if it is poisonous, radioactive, explosive, carcinogenic (causing cancer), mutagenic (causing damage to chromosomes), teratogenic (causing birth defects), or bioaccumulative (that is, increasing in concentration at the higher ends of food chains).
- Sources of toxic chemicals include improperly disposed wastewater from industrial plants and chemical process facilities (lead, mercury, chromium) as well as surface runoff containing pesticides used on agricultural areas and suburban lawns (chlordane, dieldrin, heptachlor).



https://www.nationalgeographic.com/environment/article/toxic-waste

Thermal Pollution

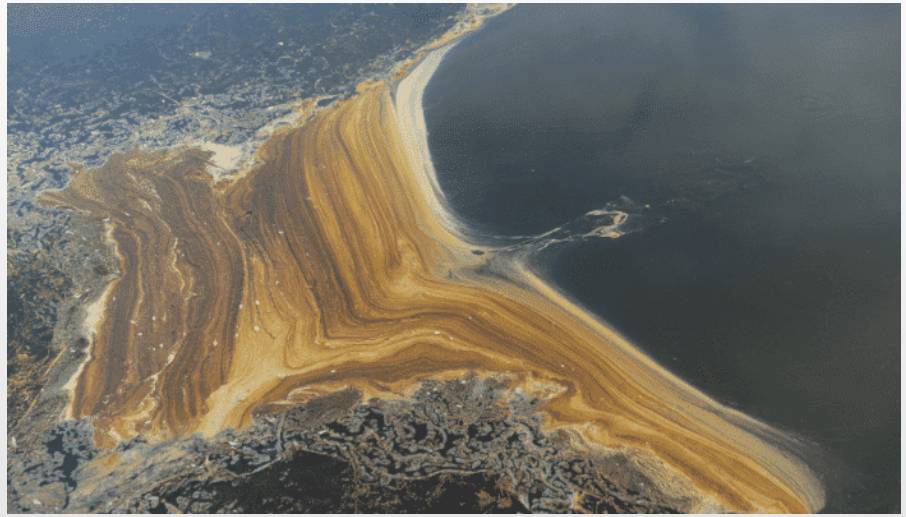
- Heat is considered to be a water pollutant because it decreases the capacity of water to hold dissolved oxygen in solution, and it increases the rate of metabolism of fish.
- Valuable species of game fish (e.g., trout) cannot survive in water with very low levels of dissolved oxygen.
- A major source of heat is the practice of discharging hot water from power plants into rivers; the discharged water may be as much as 15 °C (27 °F) warmer than the naturally occurring water.
- The rise in water temperatures because of global warming can also be considered a form of thermal pollution.



https://theberkey.com/blogs/water-filter/what-is-thermal-pollution-and-its-effects

Petroleum (oil) Pollution

- Petroleum (oil) pollution occurs when oil from roads and parking lots is carried in surface runoff into water bodies.
- Accidental oil spills are also a source of oil pollution. Oil slicks eventually move toward shore, harming aquatic life and damaging recreation areas.



https://www.marineinsight.com/environment/what-is-an-oil-spill-at-sea/



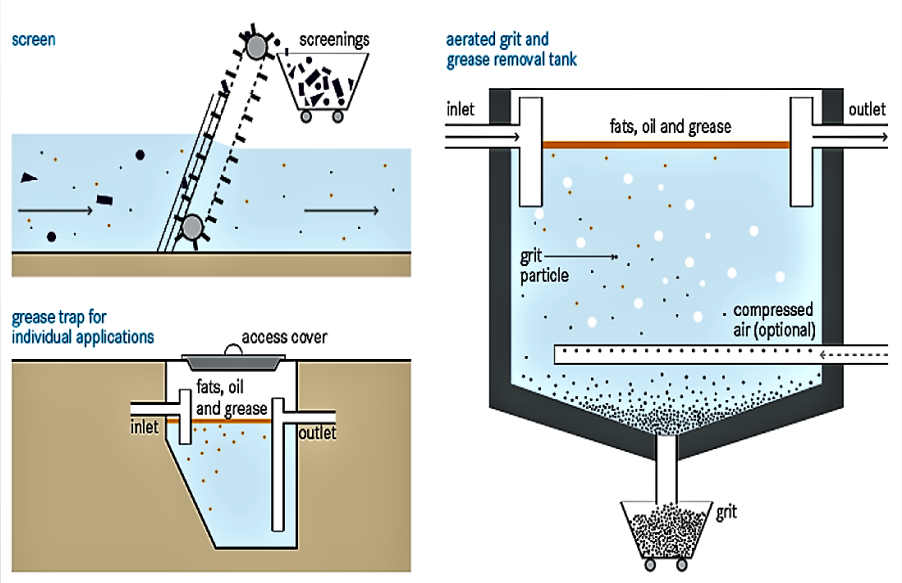
https://www.britannica.com/science/oil-spill

Water treatment (Prelimnary)

- Preliminary treatment is not as thorough nor as complete as primary, secondary, and tertiary wastewater treatment processes, so why do facilities bother with this step at all? If all contaminants and debris will be removed as part of the main treatment phases, why is preliminary treatment necessary in the first place?
- Preliminary treatment is not exclusively about protecting public health and ecosystems — this is what the main phases are designed for.
- Instead, this preliminary phase is intended to protect the water treatment equipment and infrastructure, extending their lifespan and ensuring they stay viable and effective in the long term.

Water treatment (Prelimnary)

- In a sewage treatment plant, there are tanks, filters, and other pieces of crucial equipment.
- Debris in the wastewater can rapidly erode these pieces of equipment due to friction over time or direct impact in the case of larger objects.
- As wear and tear build up over the full phase of operation, this infrastructure will need to be replaced.
- When debris is filtered and removed from the process before the main phases of treatment, the equipment is protected from harm, and the main phases of treatment become more effective.



https://sswm.info/taxonomy/term/3929/pre-treatment-technologies

Types of preliminary waste water treatment

Grit Removal

- Grit removal works similarly to screening and filtering, removing small solid particles from the wastewater mixture.
- Generally speaking, this process of removal is designed to target smaller particles of heavy solids that may be suspended in the wastewater — such as sand or grit. This is what gives the treatment process its name.
- The water is allowed to settle, and small particles with a high mass relative to the particles around them are gravity-sorted — i.e., gravity causes them to sink to the bottom of the mixture.
- Here, grit chambers are deployed to collect the particles in a specific location, where they can be safely and effectively removed.

Types of preliminary waste water treatment

Large Object Removal

- In some cases, large objects with significant mass can cause damage to filters and other treatment media, even in the preliminary phases of treatment.
- This may be achieved via a process of comminution — breaking down particles of high average size and mass into particles of lower average size and mass.
- Comminution requires high levels of energy to complete effectively, so this process is not advised in any case where it is not completely necessary.
- Alternatively, large objects can be removed from the mixture directly — perhaps caught in larger, heavier-duty grills before they reach more sensitive filtration media.

Importance of preliminary waste water treatment

- The preliminary treatment of wastewater not only ensures effective results from the main phases of the treatment process; it also helps treatment equipment and infrastructure last longer.
- In turn, treatment facilities can achieve efficient processes that reduce cost and effort while also safeguarding the long-term viability of the equipment and infrastructure.

Waste water treatment

- Once preliminary treatment is complete, the main phase of the wastewater treatment and purification can begin.
- First comes the primary wastewater treatment phase, in which solid waste within the water is sedimented into sludge and then separated from the rest of the water, ready for processing.
- Next comes the secondary stage of the wastewater treatment, in which biofiltration and oxidation are used to purify the water left behind after the primary phase of the treatment. Aeration may also be used to accelerate and optimize the process.
- In the third and final — tertiary — stage of treatment, phosphates and nitrates are removed from the remaining wastewater, neutralizing the effluent and protecting local communities and ecosystems.
- The success of each of these three main stages — as well as the longevity of the equipment — depends upon the successful completion of the preliminary processes.

Primary water treatment

- Primary treatment removes material that will either float or readily settle out by gravity.
- It includes the physical processes of screening, comminution, grit removal, and sedimentation. Screens are made of long, closely spaced, narrow metal bars.
- They block floating debris such as wood, rags, and other bulky objects that could clog pipes or pumps. In modern plants the screens are cleaned mechanically, and the material is promptly disposed of by burial on the plant grounds.
- A comminutor may be used to grind and shred debris that passes through the screens. The shredded material is removed later by sedimentation or flotation processes.
- Grit chambers are long narrow tanks that are designed to slow down the flow so that solids such as sand, coffee grounds, and eggshells will settle out of the water.

https://www.britannica.com/science/biogeochemical-cycle

Primary water treatment

- Grit causes excessive wear and tear on pumps and other plant equipment. Its removal is particularly important in cities with combined sewer systems, which carry a good deal of silt, sand, and gravel that wash off streets or land during a storm.
- Suspended solids that pass through screens and grit chambers are removed from the sewage in sedimentation tanks.
- These tanks, also called primary clarifiers, provide about two hours of detention time for gravity settling to take place.
- As the sewage flows through them slowly, the solids gradually sink to the bottom. The settled solids—known as raw or primary sludge—are moved along the tank bottom by mechanical scrapers.
- Sludge is collected in a hopper, where it is pumped out for removal.
- Mechanical surface-skimming devices remove grease and other floating materials.

Secondary Treatment

- Secondary treatment removes the soluble organic matter that escapes primary treatment. It also removes more of the suspended solids.
- Removal is usually accomplished by biological processes in which microbes consume the organic impurities as food, converting them into carbon dioxide, water, and energy for their own growth and reproduction.
- The sewage treatment plant provides a suitable environment, albeit of steel and concrete, for this natural biological process.
- Removal of soluble organic matter at the treatment plant helps to protect the dissolved oxygen balance of a receiving stream, river, or lake.
- There are three basic biological treatment methods: the trickling filter, the activated sludge process, and the oxidation pond.
- A fourth, less common method is the rotating biological contacter.

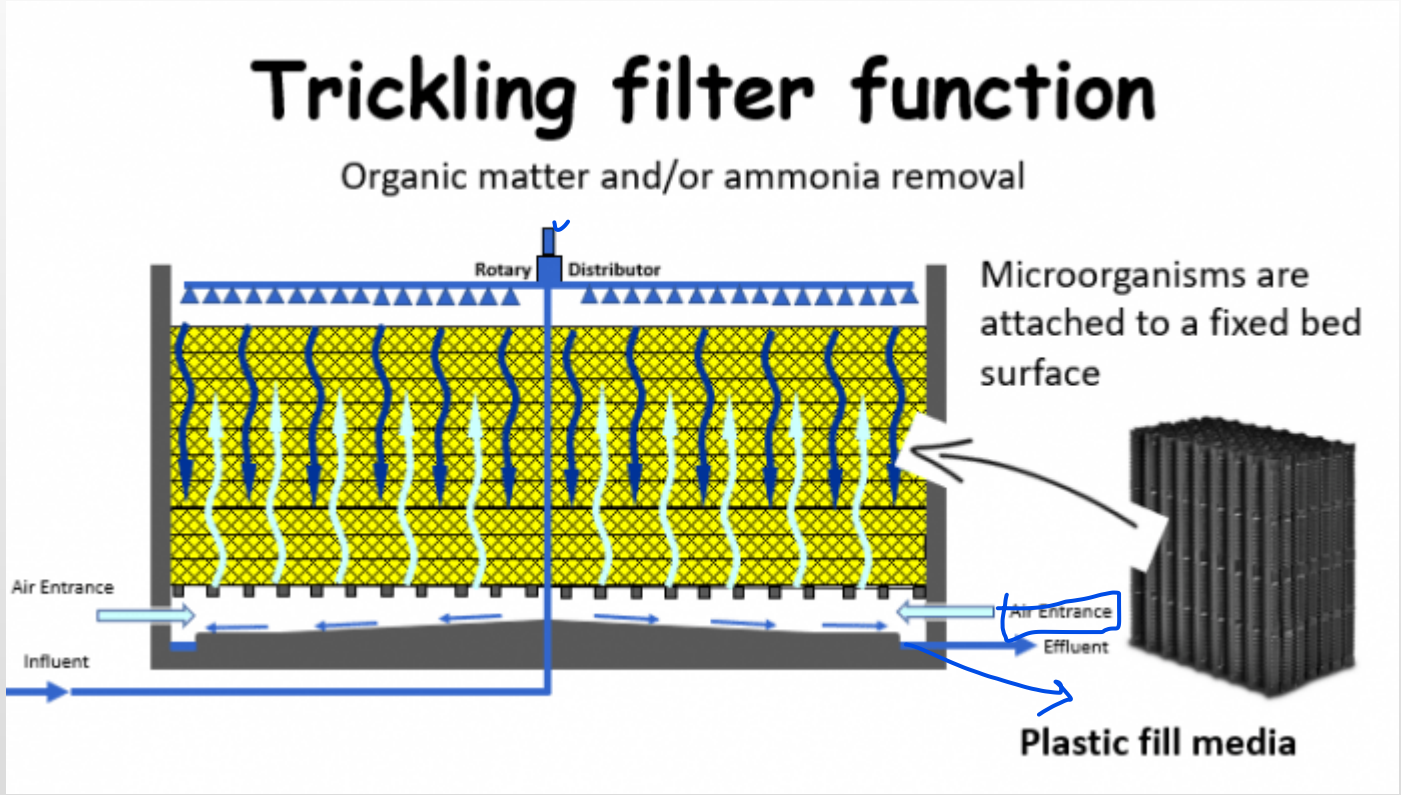
Secondary Treatment

Trickling filter

- A trickling filter is simply a tank filled with a deep bed of stones.
- Settled sewage is sprayed continuously over the top of the stones and trickles to the bottom, where it is collected for further treatment.
- As the wastewater trickles down, bacteria gather and multiply on the stones.
- The steady flow of sewage over these growths allows the microbes to absorb the dissolved organics, thus lowering the biochemical oxygen demand (BOD) of the sewage. Air circulating upward through the spaces among the stones provides sufficient oxygen for the metabolic processes.

Settling tanks, called secondary clarifiers, follow the trickling filters. These clarifiers remove microbes that are washed off the rocks by the flow of wastewater. Two or more trickling filters may be connected in series, and sewage can be recirculated in order to increase treatment efficiencies.

Secondary Treatment



https://smartwatermagazine.com/blogs/karl-uwe-schmitz/trickling-filter-design-guideline-how-do-trickling-filters-work

Secondary Treatment

Activated sludge

- The activated sludge treatment system consists of an aeration tank followed by a secondary clarifier.
- Settled sewage, mixed with fresh sludge that is recirculated from the secondary clarifier, is introduced into the aeration tank.
- Compressed air is then injected into the mixture through porous diffusers located at the bottom of the tank.
- As it bubbles to the surface, the diffused air provides oxygen and a rapid mixing action. Air can also be added by the churning action of mechanical propeller-like mixers located at the tank surface.
- Under such oxygenated conditions, microorganisms thrive, forming an active, healthy suspension of biological solids—mostly bacteria—called activated sludge.

Secondary Treatment

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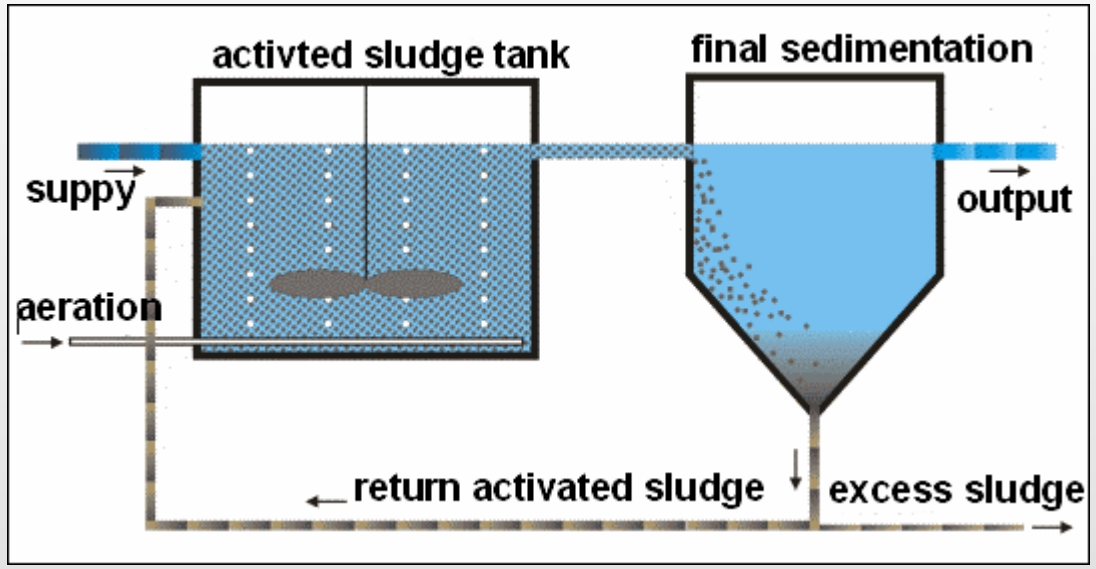
Activated sludge

- About six hours of detention is provided in the aeration tank.
- This gives the microbes enough time to absorb dissolved organics from the sewage, reducing the BOD.
- The mixture then flows from the aeration tank into the secondary clarifier, where activated sludge settles out by gravity.
- Clear water is skimmed from the surface of the clarifier, disinfected, and discharged as secondary effluent.
- The sludge is pumped out from a hopper at the bottom of the tank.
- About 30 percent of the sludge is recirculated back into the aeration tank, where it is mixed with the primary effluent. This recirculation is a key feature of the activated sludge process.
- The recycled microbes are well acclimated to the sewage environment and readily metabolize the organic materials in the primary effluent. The remaining 70 percent of the secondary sludge must be treated and disposed of in an acceptable manner (*see* Sludge treatment and disposal).

Secondary Treatment

Activated sludge

- The recycled microbes are well acclimated to the sewage environment and readily metabolize the organic materials in the primary effluent.
- The remaining 70 percent of the secondary sludge must be treated and disposed of in an acceptable manner (*see* Sludge treatment and disposal).



https://cgi.tuharburg.de/~awwweb/wbt/emwater/lessons/lesson_c1/lm_pg_1425.html

Secondary Treatment

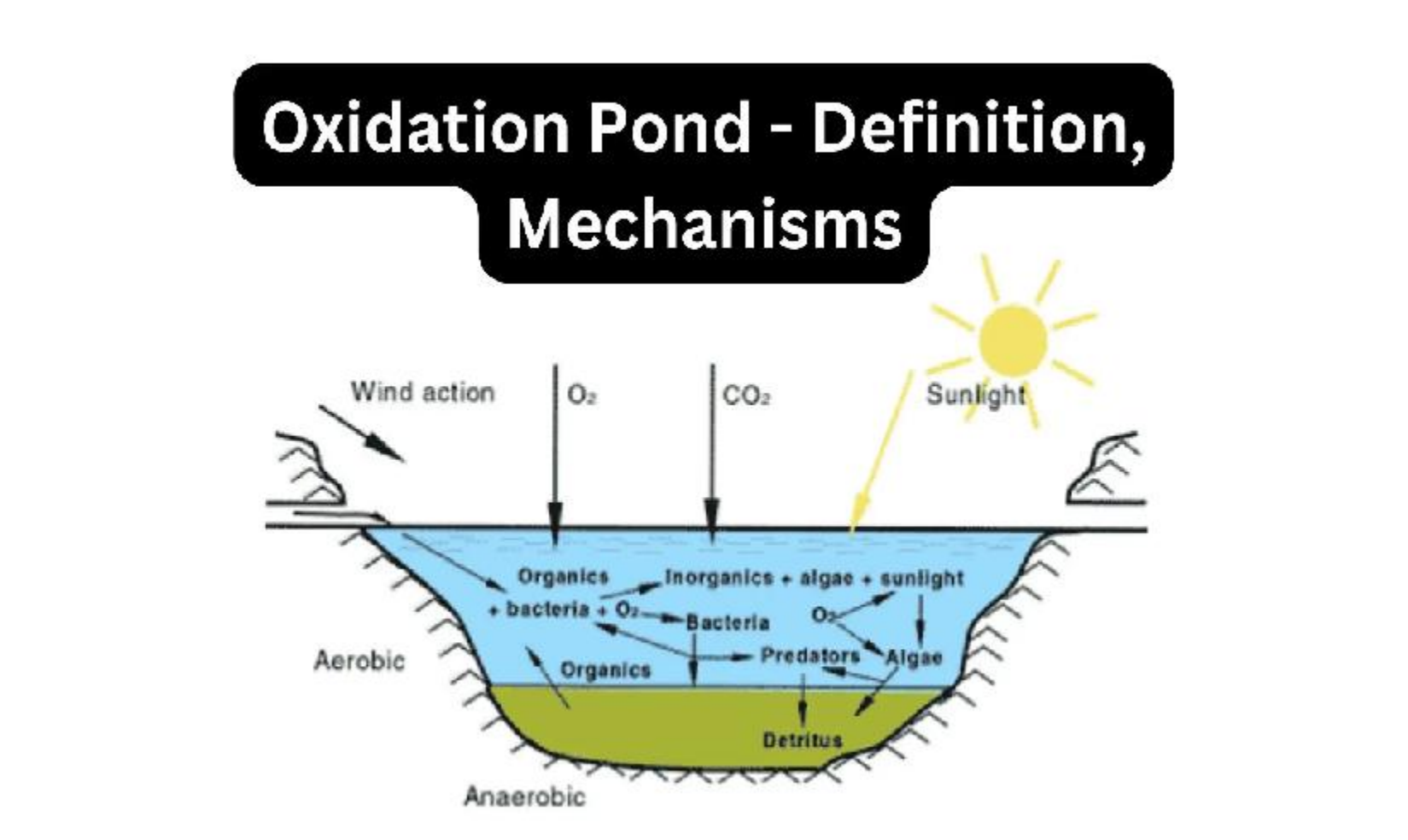
- Variations of the activated sludge process include extended aeration, contact stabilization, and high-purity oxygen aeration.
- Extended aeration and contact stabilization systems omit the primary settling step.
- They are efficient for treating small sewage flows from motels, schools, and other relatively isolated wastewater sources.
- Both of these treatments are usually provided in prefabricated steel tanks called package plants.
- Oxygen aeration systems mix pure oxygen with activated sludge.
- A richer concentration of oxygen allows the aeration time to be shortened from six to two hours, reducing the required tank volume.

Secondary Treatment

Oxidation pond

- Oxidation ponds, also called lagoons or stabilization ponds, are large, shallow ponds designed to treat wastewater through the interaction of sunlight, bacteria, and algae.
- Algae grow using energy from the sun and carbon dioxide and inorganic compounds released by bacteria in water.
- During the process of photosynthesis, the algae release oxygen needed by aerobic bacteria.
- Mechanical aerators are sometimes installed to supply yet more oxygen, thereby reducing the required size of the pond.
- Sludge deposits in the pond must eventually be removed by dredging/scooping.
- Algae remaining in the pond effluent can be removed by filtration or by a combination of chemical treatment and settling.

Secondary Treatment

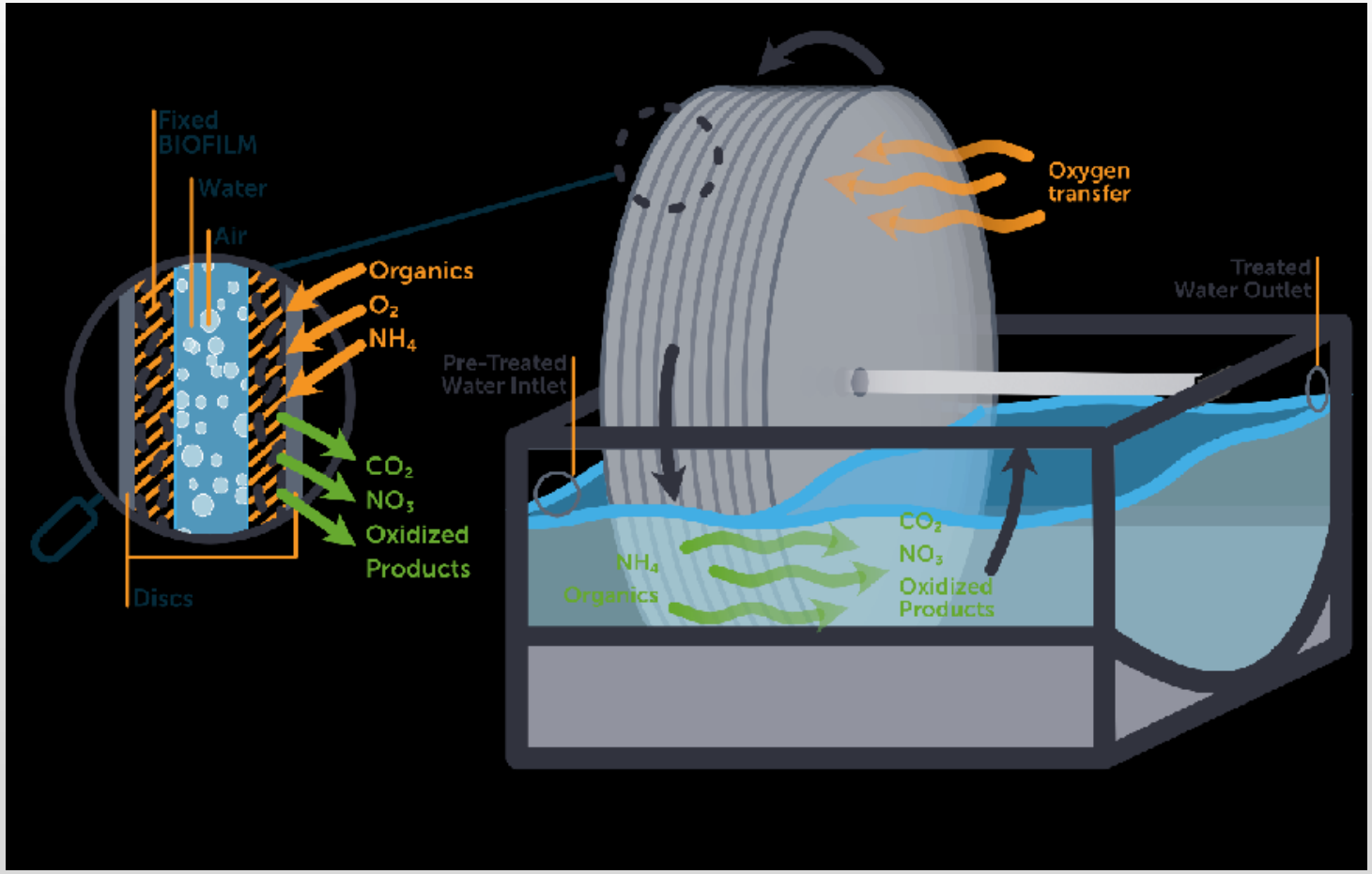


https://microbiologynote.com/oxidation-pond/

Secondary Treatment

Rotating biological contacter

➢In this treatment system a series of large plastic disks mounted on a horizontal shaft are partially submerged in primary effluent. As the shaft rotates, the disks are exposed alternately to air and wastewater, allowing a layer of bacteria to grow on the disks and to metabolize the organics in the wastewater.



https://www.biorotor.com/how-biorotor-works/

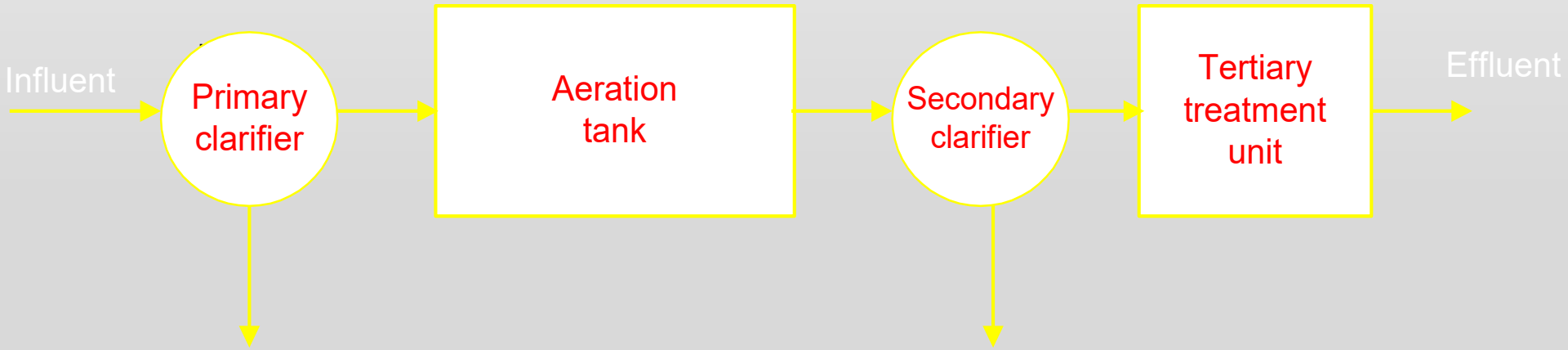
- Introduction
- Tertiary Treatment Technologies
- Removal of Residual Constituents
  - Suspended Solids Removal
  - Nutrients Removal
  - Removal of Toxic Compounds
  - Removal of Dissolved Inorganic Compounds
- Tertiary Treatment of Wastewater in Saudi Arabia
- Summary

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INTRODUCTION

- **Definition**

Further removal of suspended and dissolved contaminants, not normally removed by conventional treatment



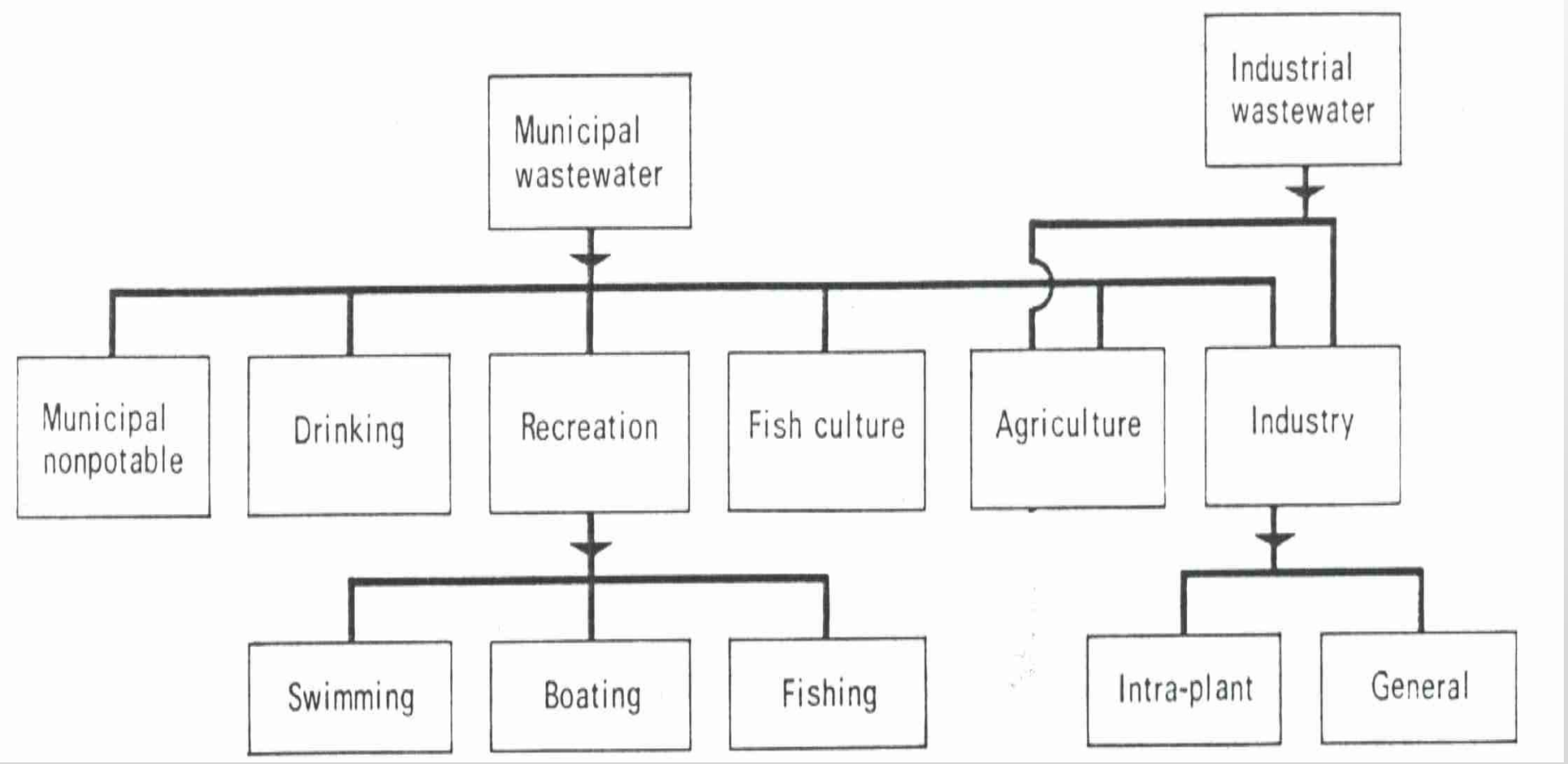
- **Need of Tertiary Treatment of Wastewater:**
  - **Continued increase in population**
  - **Limited water resources**
  - **Contamination of both surface and groundwater**
  - **Uneven distribution of water resources and**
  - **periodic draughts**

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- **Typical Constituents Present in Wastewater:**
  - **Suspended solids**
  - **Biodegradable organics compounds**
  - **Volatile organic compounds**
  - **Toxic contaminants**
  - **Nutrients**
  - **Other organics and inorganics**

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- Options for reuse of treated wastewater



TERTIARY TREATMENT TECHNOLOGIES

- **Classification of Technologies**
  - **Primary Treatment Systems**
  - **Secondary Treatment Systems**
  - **Tertiary Treatment Systems**

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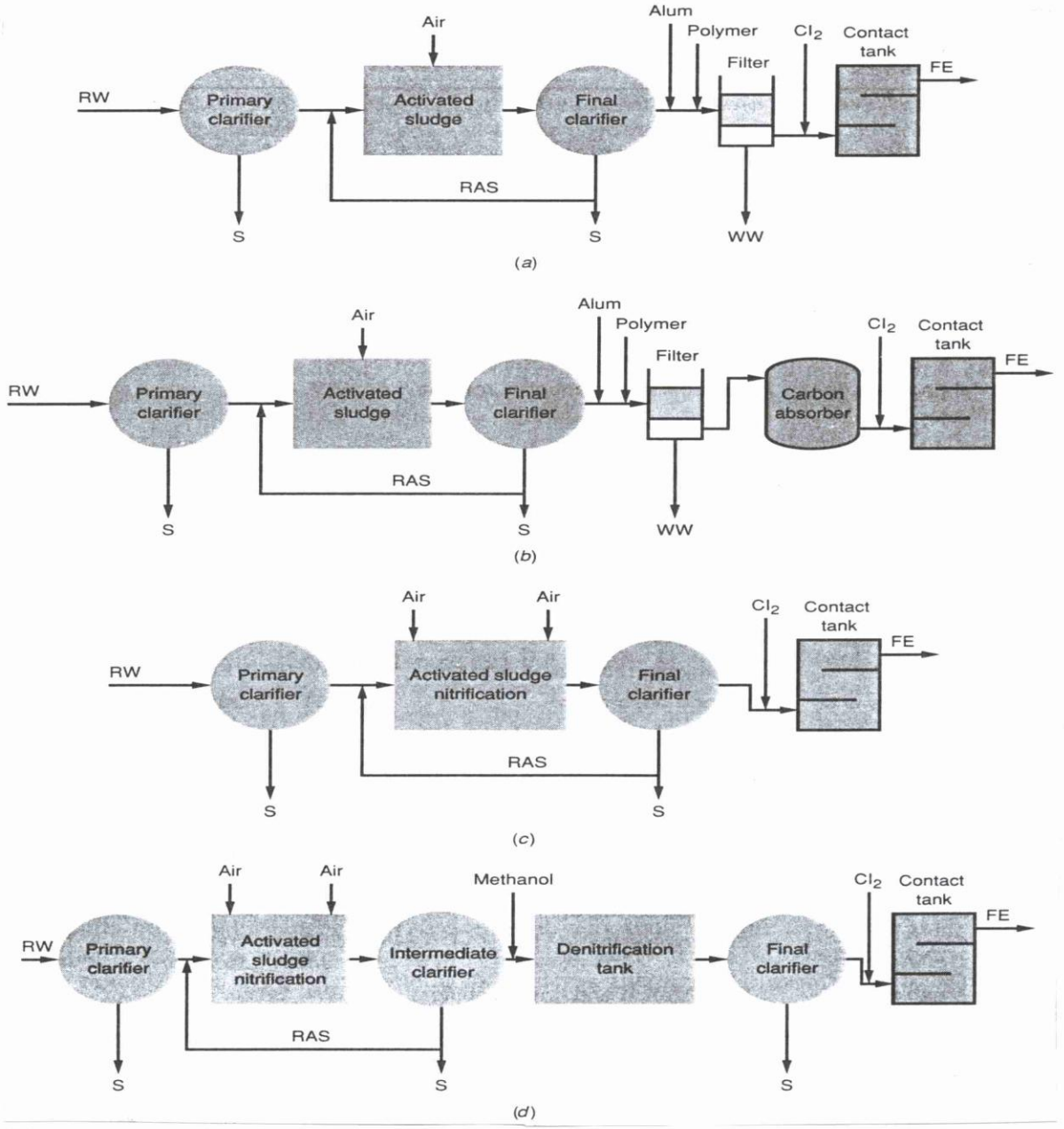


Fig. 2. Examples of tertiary treatment flow diagrams: (a) activated sludge + filtration, (b) activated sludge + filtration + activated carbon, (c) activated sludge nitrification (single stage), and (d) activated sludge nitrification/denitrification using methanol

- **Factors affecting the selection of treatment processes:**
  - **The potential use of the treated effluent**
  - **The nature of the wastewater**
  - **The compatibility of the various operations and processes**
  - **The available means to dispose of the ultimate contaminants, and**
  - **The environmental and economic feasibility of the various systems**

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Typical performance data:

Table 4.0:

Treatment levels achievable with various combinations of unit operations and processes used for advanced wastewater treatment[a]

Treatment process	Typical effluent quality						
	SS, mg/L	BOD_5, mg/L	COD, mg/L	Total N, mg/L	NH_3-N, mg/L	PO_4 as P, mg/L	Turbidity, NTU
Activated sludge + granular-medium filtration	4–6	<5–10	30–70	15–35	15–25	4–10	0.3–5
Activated sludge + granular-medium filtration + carbon adsorption	<3	<1	5–15	15–30	15–25	4–10	0.3–3
Activated sludge/nitrification, single stage	10–25	5–15	20–45	20–30	1–5	6–10	5–15
Activated sludge/nitrification-denitrification, separate stages	10–25	5–15	20–35	5–10	1–2	6–10	5–15
Metal salt addition to activated sludge	10–20	10–20	30–70	15–30	15–25	<2	5–10
Metal salt addition to activated sludge + nitrification/denitrification + filtration	<5–10	<5–10	20–30	3–5	1–2	<1	0.3–3
Mainstream biological phosphorus removal[b]	10–20	5–15	20–35	15–25	5–10	<2	5–10
Mainstream biological nitrogen and phosphorus removal[b] + filtration	<10	<5	20–30	<5	<2	<1	0.3–3

- **What are the contaminants removed during tertiary treatment?**
  - **Suspended solids**
  - **Nutrients**
  - **Toxic compounds**
  - **Dissolved organics and inorganics**

REMOVAL OF RESIDUAL CONSTITUNTS

- **Suspended Solids Removal:**
  - **Granular-medium filters**
    - **the bed depth**
    - **the type of filtering medium used**
    - **whether the filtering medium is stratified or unstratified**
    - **the type of operation**
  - **Microstrainers**

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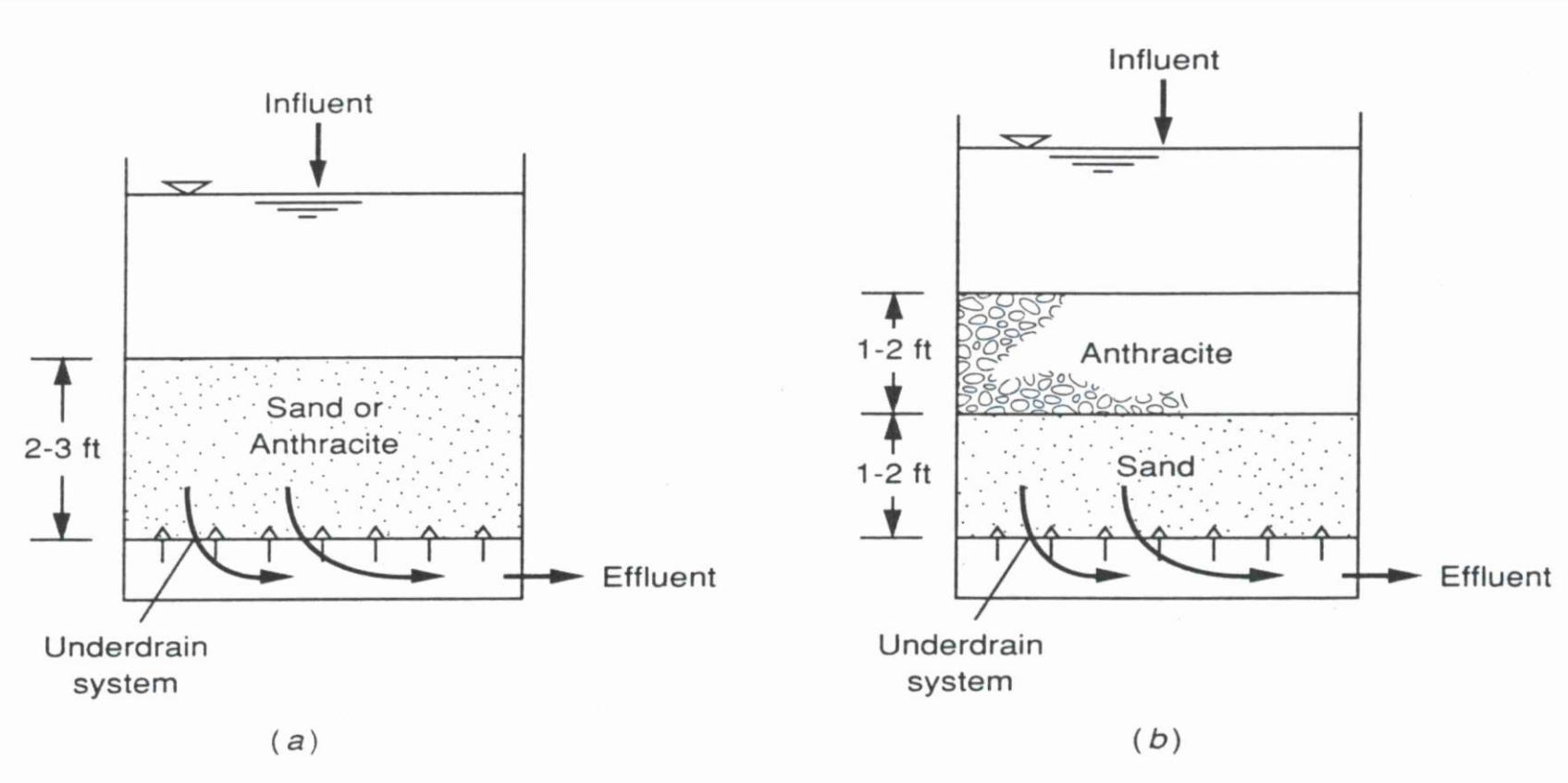


Fig. 12a. Types of shallow-bed filters (a) mono-medium downflow, (b) dual-medium downflow

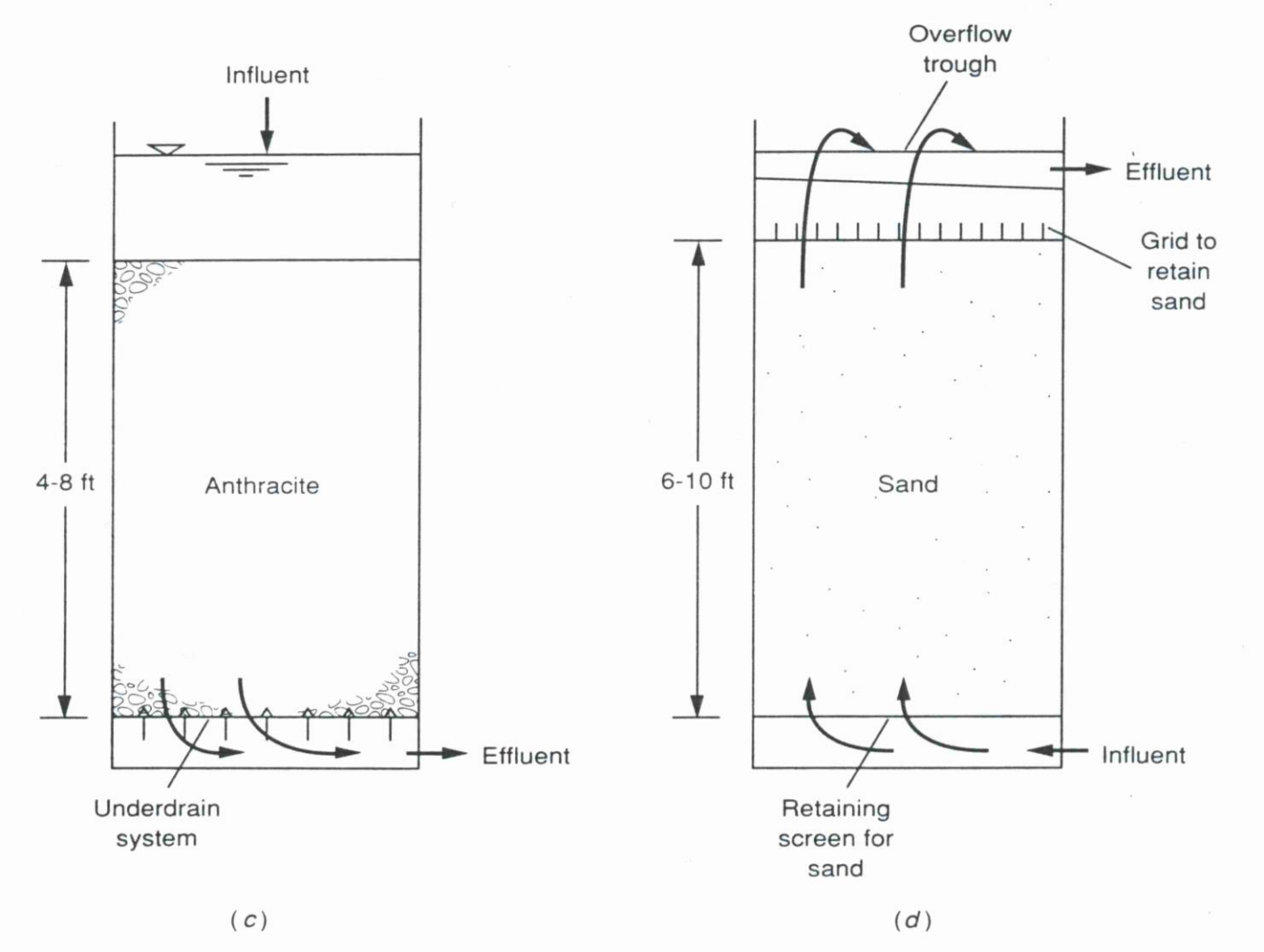


Fig. 12b. Types of deep-bed filters (a) mono-medium downflow, (b) mono-medium upflow

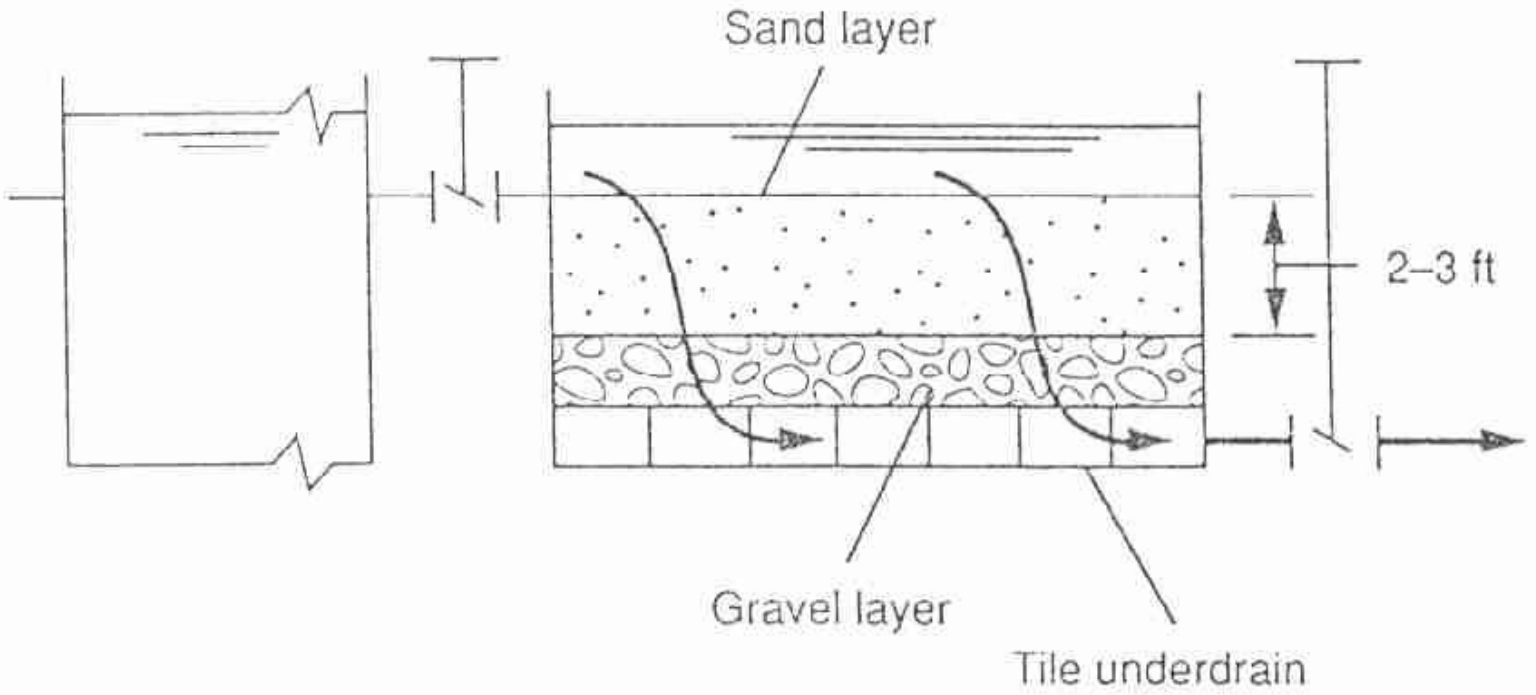


Fig. 6. Typical slow sand filter

Table 6.0:

Typical design data for mono-medium filters

Characteristic	Value	
	Range	Typical
Shallow-bed (stratified)		
Sand:		
Depth, in	10–12	11
Effective size, mm	0.35–0.6	0.45
Uniformity coefficient	1.2–1.6	1.5
Filtration rate, gal/ft^2 · min	2–6	3
Anthracite:		
Depth, in	12–20	16
Effective size, mm	0.8–1.5	1.3
Uniformity coefficient	1.3–1.8	1.6
Filtration rate, gal/ft^2 · min	2–6	3
Conventional (stratified)		
Sand:		
Depth, in	20–30	24
Effective size, mm	0.4–0.8	0.65
Uniformity coefficient	1.2–1.6	1.5
Filtration rate, gal/ft^2 · min	2–6	3
Anthracite:		
Depth, in	24–36	30
Effective size, mm	0.8–2.0	1.3
Uniformity coefficient	1.3–1.8	1.6
Filtration rate, gal/ft^2 · min	2–8	4
Deep-bed (unstratified)		
Sand:		
Depth, in	36–72	48
Effective size, mm	2–3	2.5
Uniformity coefficient	1.2–1.6	1.5
Filtration rate, gal/ft^2 · min	2–10	5
Anthracite:		
Depth, in	36–84	60
Effective size, mm	2–4	2.75
Uniformity coefficient	1.3–1.8	1.6
Filtration rate, gal/ft^2 · min	2–10	5

Table 7.0:

Typical design data for dual- and multi-medium filters[a]

Characteristic	Value	
	Range	Typical
Dual-medium		
Anthracite:		
Depth, in	12–30	24
Effective size, mm	0.8–2.0	1.3
Uniformity coefficient	1.3–1.8	1.6
Sand:		
Depth, in	6–12	12
Effective size, mm	0.4–0.8	0.65
Uniformity coefficient	1.2–1.6	1.5
Filtration rate, gal/ft^2 · min	2–10	5
Multimedium		
Anthracite (top layer of quad-media filter):		
Depth, in	8–20	16
Effective size, mm	1.3–2.0	1.6
Uniformity coefficient	1.5–1.8	1.6
Anthracite (second layer of quad-media filter):		
Depth, in	4–16	8
Effective size, mm	1.0–1.6	1.1
Uniformity coefficient	1.5–1.8	1.6
Anthracite (top layer of tri-media filter):		
Depth, in	8–20	16
Effective size, mm	1.0–2.0	1.4
Uniformity coefficient	1.4–1.8	1.6
Sand:		
Depth, in	8–16	10
Effective size, mm	0.4–0.8	0.5
Uniformity coefficient	1.3–1.8	1.6
Garnet or ilmenite:		
Depth, in	2–6	4
Effective size, mm	0.2–0.6	0.3
Uniformity coefficient	1.5–1.8	1.6
Filtration rate, gal/ft^2 · min	2–10	5

- **Sizing of a filter:**
  - **Principal design criteria of a filter design is water flow rate and head loss**
  - **Usually we know**
    - **flow rate of influent**
    - **surface loading rate**

$$\text{Surface area of filtering unit} = \frac{\text{Flow rate}}{\text{Surface loading rate}}$$

- Head loss can be calculated using reference (MetCalf & Eddy, 1991)

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- Removal of Suspended Solids by Microscreaning:

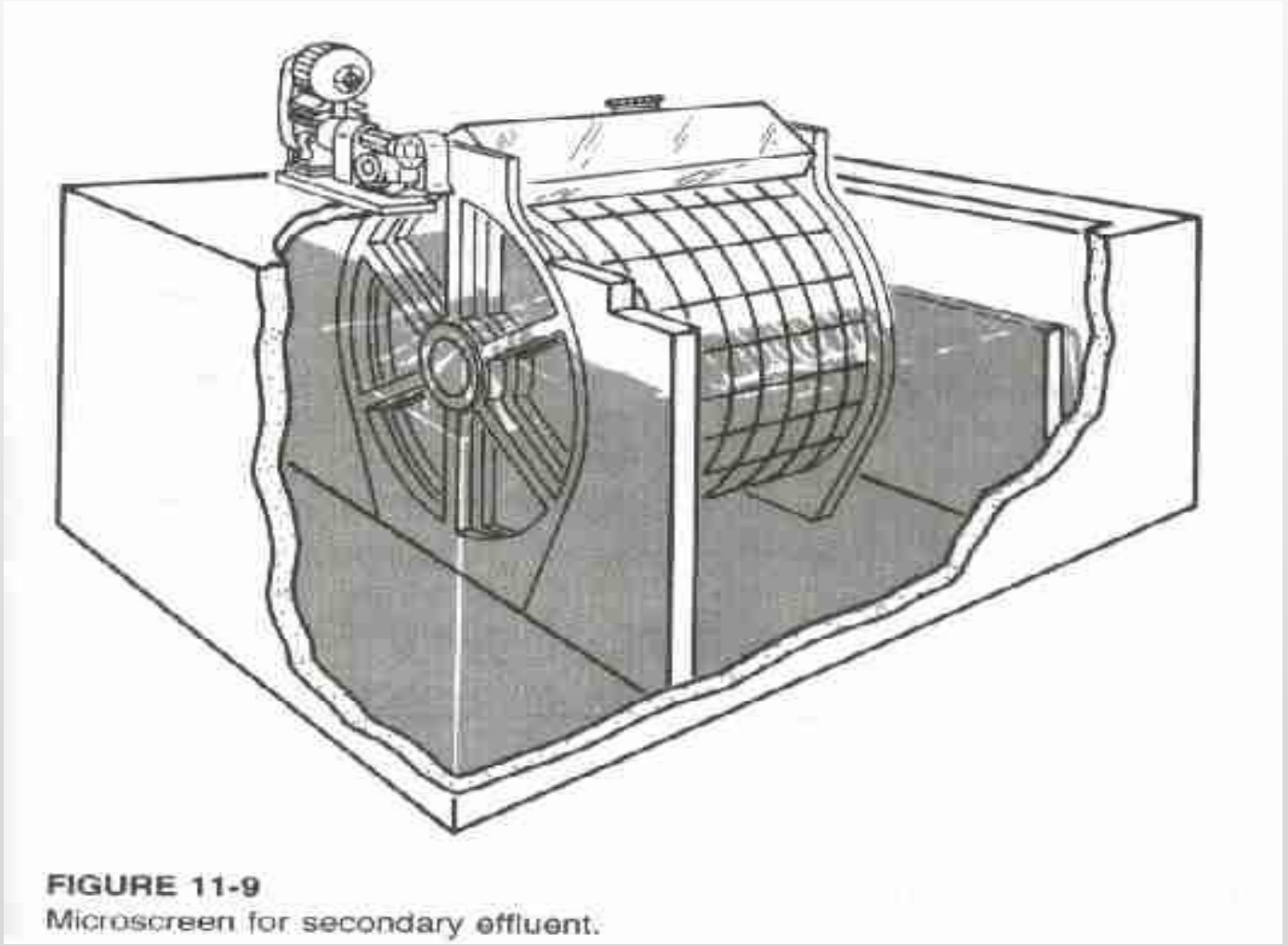


FIGURE 11-9
Microscreen for secondary effluent.

Table 9.0:

Typical design information for microscreens used for screening secondary settled effluent[a]

Item	Typical value	Remarks
Screen size	20 – 35 μm	Stainless steel or polyester screen cloths are available in sizes ranging from 15 to 60 μm
Hydraulic loading rate	75 – 150 gal/ft^2 · min	Based on submerged surface area of drum
Headloss through screen	3 – 6 in	Bypass should be provided when headloss exceeds 8 in
Drum submergence	70 – 75% of height; 60 – 70% of area	Varies depending on screen design
Drum diameter	8 – 16 ft	10 ft (3 m) is most commonly used size; smaller sizes increase backwash requirements
Drum speed	15 ft/min at 3 in headloss; 115 – 150 ft/min at 6 in headloss	Maximum rotational speed is limited to 150 ft/min
Backwash requirements	2% of throughput at 50 lb$_f$/in^2; 5% of throughput at 15 lb$_f$/in^2	

- (II) Nutrients Removal

  Basic nutrients present in the domestic wastewater are

  - Nitrogen (ammonia, nitrite, nitrate)
  - Phosphorus (soluble and insoluble)
  - Sulfate
  - Other compounds of nitrogen & phosphorus

  Problems associated with nutrients presence in wastewater are

  - accelerate the eutrophication
  - stimulate the growth of algae & rooted aquatic plants
  - aesthetic problems & nuisance

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- depleting D.O. concentration in receiving waters
- Toxicity towards aquatic life
- increasing chlorine demand
- presenting a public health hazard
- affecting the suitability of wastewater for reuse

- Nutrient Control could be accomplished by:
  - physical methods
  - chemical methods, and
  - biological methods

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Control and Removal of Nitrogen (Biologically):

- Removal of Nitrogen by Nitrification/Denitrification Processes:
  - It is a two step processes

$$NH_4^- \xrightarrow{\text{aerobic}} NO_3^- \text{ (nitrification)}$$

$$NO_3^- \xrightarrow{\text{anoxic}} N_2 \text{ (denitrification)}$$

- Removal of Nitrogen by Nitrification Processes:
  - 1) Single-stage process
  - 2) Separate-stage process



Fig. 8a. Typical carbon oxidation and nitrification processes (single-stage)

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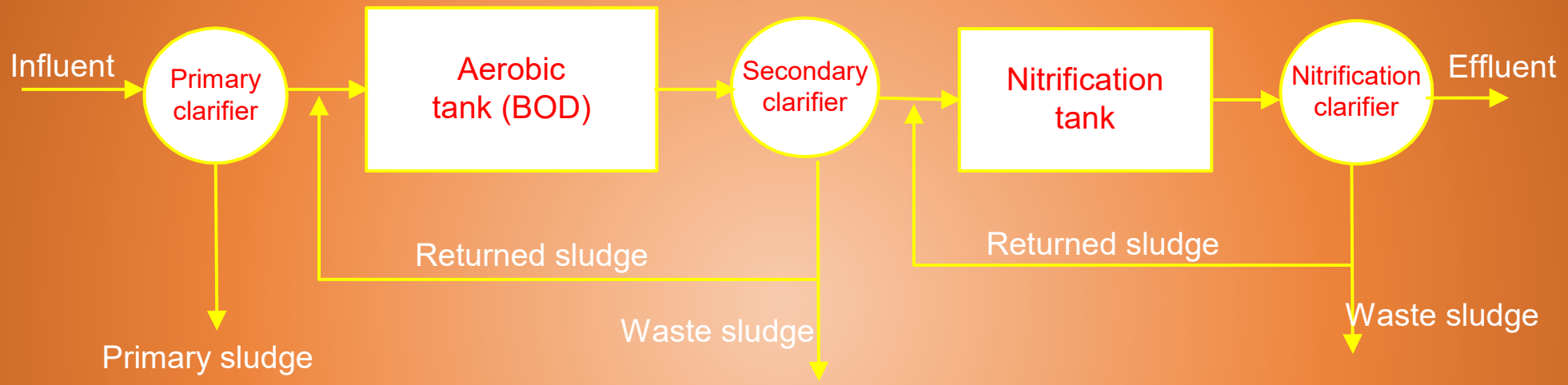


Fig. 8b. Typical carbon oxidation and nitrification processes (separate-stage)

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Table 10.0:

Comparison of nitrification alternatives

System type	Advantages	Disadvantages
Combined carbon oxidation nitrification		
Suspended-growth	Combined treatment of carbon and ammonia in a single stage; low effluent ammonia is possible; inventory control of mixed-liquor stable due to high BOD_5/TKN ratio	No protection against toxicants; only moderate stability of operation; stability linked to operation of secondary clarifier for biomass return; large reactors required in cold weather
Attached-growth	Combined treatment of carbon and ammonia in a single stage; stability not linked to secondary clarifier as organisms are attached to media	No protection against toxicants; only moderate stability of operation; effluent ammonia is normally 1–3 mg/L (except RBC); cold weather operation impractical in most cases
Separate-stage nitrification		
Suspended-growth	Good protection against most toxicants; stable operation; low effluent ammonia possible	Sludge inventory requires careful control when BOD_5/TKN ratio is low; stability of operation linked to operation of secondary clarifier for biomass return; greater number of unit processes required than for combined carbon oxidation nitrification
Attached-growth	Good protection against most toxicants; stable operation; stability not linked to secondary clarifier as organisms are attached to media	Effluent ammonia normally 1–3 mg/L; greater number of unit processes required than for combined carbon oxidation nitrification

- Nitrification/Denitrification systems can be classified as:

  (a) Combined Nitrification/Denitrification Systems

  1) Bardenpho process(four stage)

  2) Oxidation Ditch process

  (b) Separate-Stage Denitrification Systems

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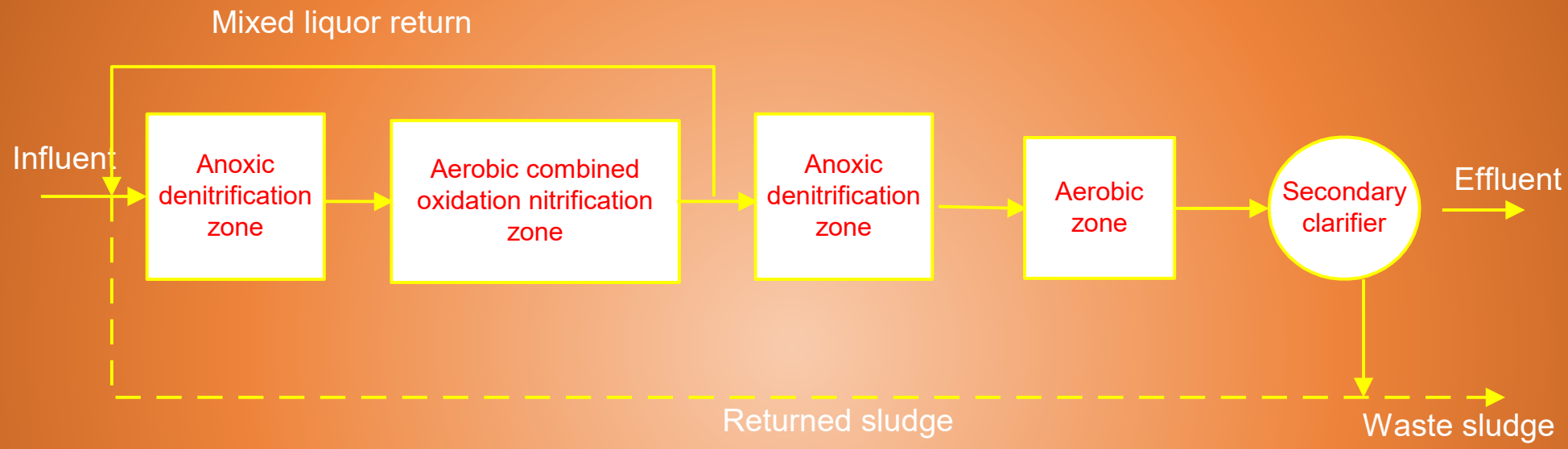


Fig. 9a. Combined-stage nitrification/denitrification system (four-stage Bardenpho)

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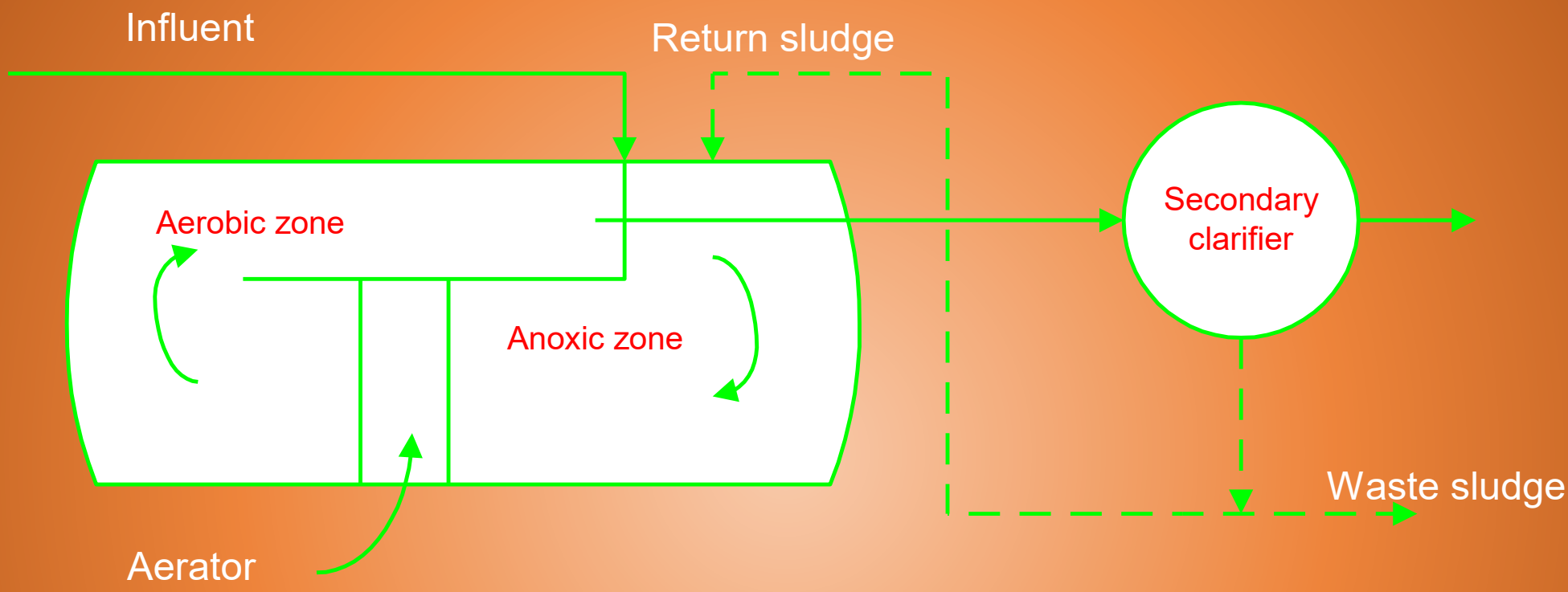


Fig. 9b. Combined-stage nitrification/denitrification system (oxidation ditch)

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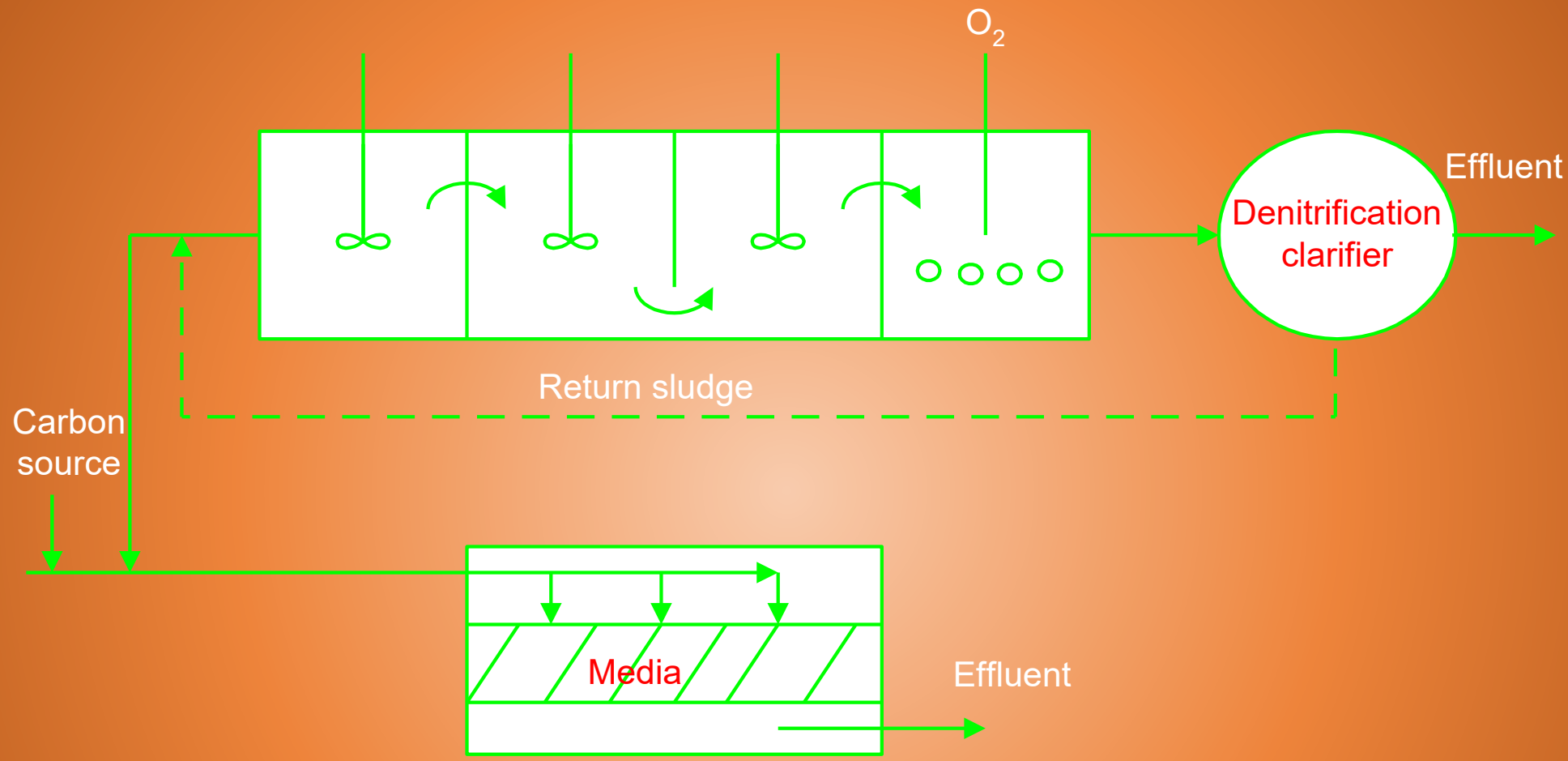


Fig. 10. Separate-stage denitrification process using a separate carbon source

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- Control and Removal of Nitrogen (Physical & Chemical Methods):
  - air Stripping
  - breakpoint chlorination
  - selective ion exchange

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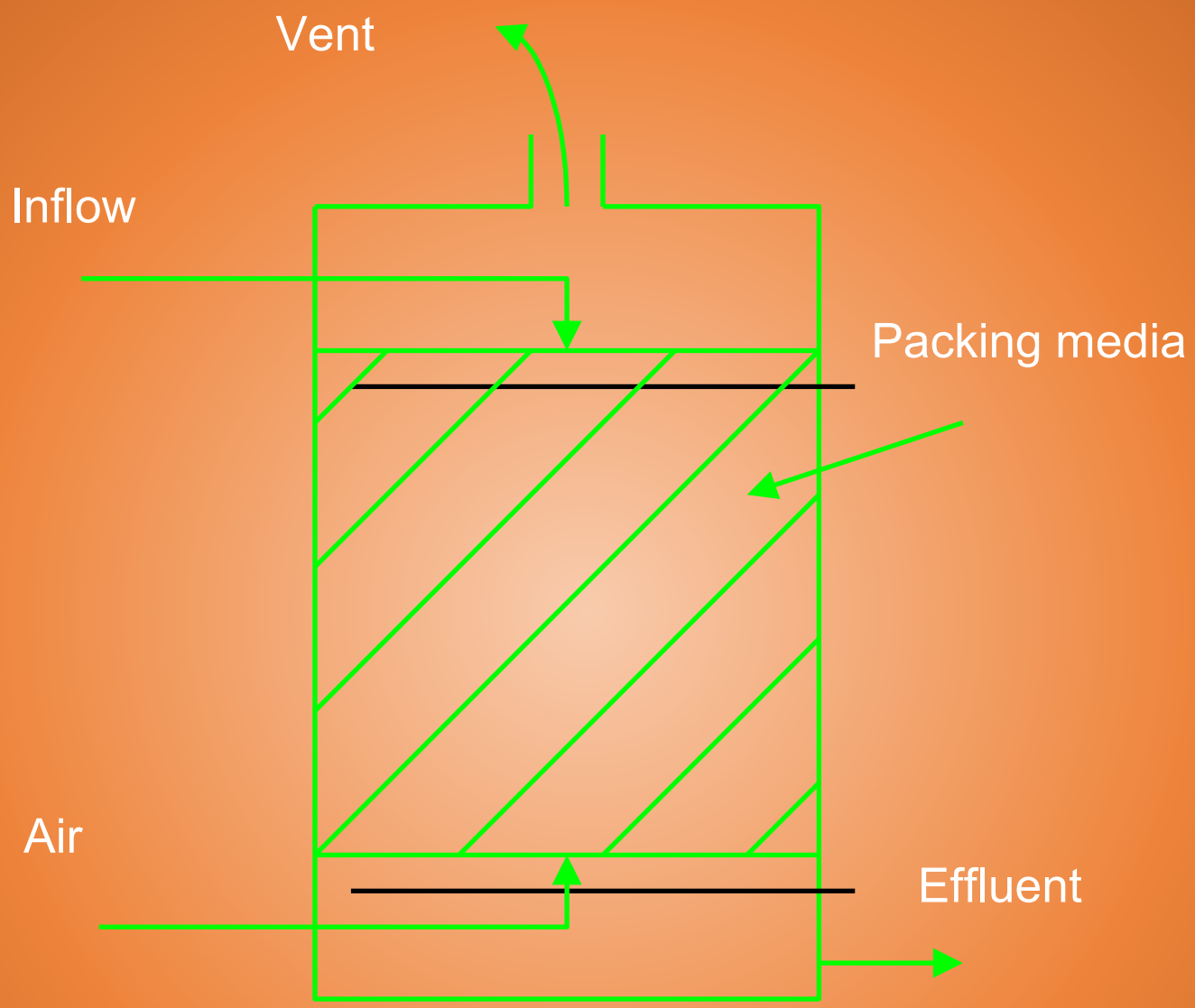


Fig. 11. Cross-section of a countercurrent ammonia-stripping tower

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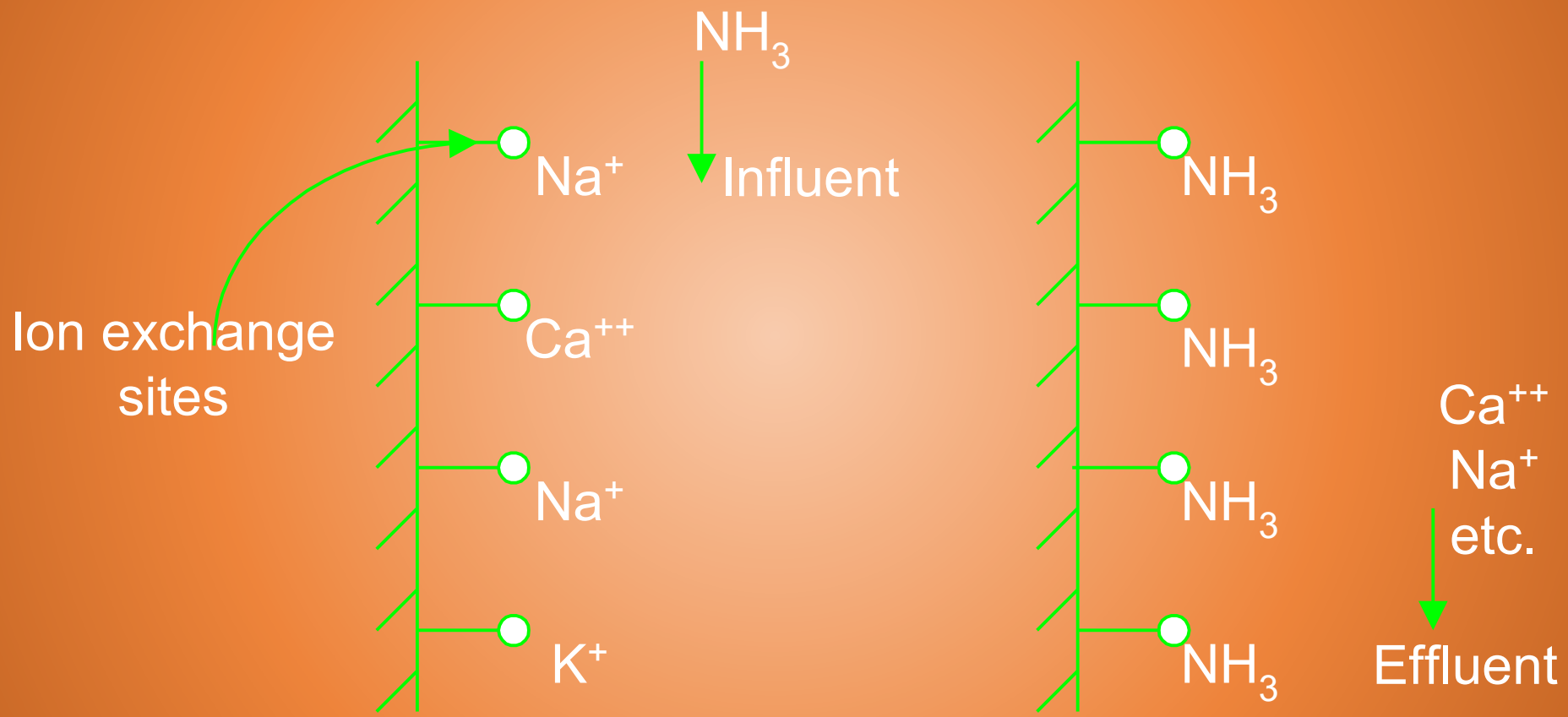
- Breakpoint chlorination:
  - Oxidation of ammonia-nitrogen can be done by adding excess chlorine
  - Basic chemical equations:

$$Cl_2 + H_2O \rightarrow HOCl + H^+ + Cl^-$$

$$NH_3 + HOCl \rightarrow N_2\uparrow + N_2O\uparrow + NO_2^- + NO_3^- + Cl^-$$

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- Ion exchange process



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Table 12.0: Advantaged and disadvantages of physical and chemical nitrogen removal processes

Process	Advantages	Disadvantages
Air stripping	Process can be controlled for selected ammonia removals. Most applicable if required seasonally in combination with lime system for phosphorus removal. Process may be able to meet total nitrogen standards. Not sensitive to toxic substances.	Process is temperature sensitive. Ammonia solubility increases with lower temperatures. Air requirements also vary. Fogging and icing occur in cold weather. Ammonia reaction with sulfur dioxide may cause air pollution problems. Process usually requires lime for pH control, thereby increasing treatment cost and lime-related operating and maintenance problems. Carbonate scaling of packing and piping. Potential noise and aesthetic problems.
Breakpoint chlorination	With proper control, all ammonia nitrogen can be oxidized. Process can be used following other nitrogen removal processes for fine-tuning of nitrogen removal. Concurrent effluent disinfection. Limited space requirement. Not sensitive to toxic substances and temperature. Low capital costs. Adaptable to existing facility.	May produce high chlorine residuals that are toxic to aquatic organisms. Wastewater contains a variety of chlorine demanding substances which increase cost of treatment. Process is sensitive to pH, which affects dosage requirements. High operating cost due to chemical requirements. Trihalomethane formation may impact quality of water supplies. Addition of chlorine raises effluent TDS. Process may not be able to meet total nitrogen standards. Requires careful control of pH to avoid formation of nitrogen trichloride gas. Requires highly skilled operator.
Ion exchange	Can be used where climatic conditions inhibit biological nitrification and where stringent effluent standards are required. Produces a relatively low TDS effluent. Produces a reclaimable product (aqueous ammonia). Process may be able to meet total nitrogen standards. Ease of product quality control.,	Organic matter in effluent from biological treatment can cause resin binding. Pretreatment by filtration is usually required to prevent the buildup of excessive headloss due to suspended-solids accumulation. High concentrations of other cations will reduce ammonia removal capability. Regeneration recovery may require the addition of another unit progress (e.g., gas stripping). High capital and operating costs. Regeneration products must be disposed of. Requires highly skilled operator.

Phosphorus Removal

Biologically:

Key to the biological phosphorus removal is the exposure of the microorganisms to alternating anaerobic & aerobic conditions

- Phosphorus Removal Processes
  - (1) Mainstream process
  - (2) Sidestream process
  - (3) Sequencing Batch Reactor (SBR)

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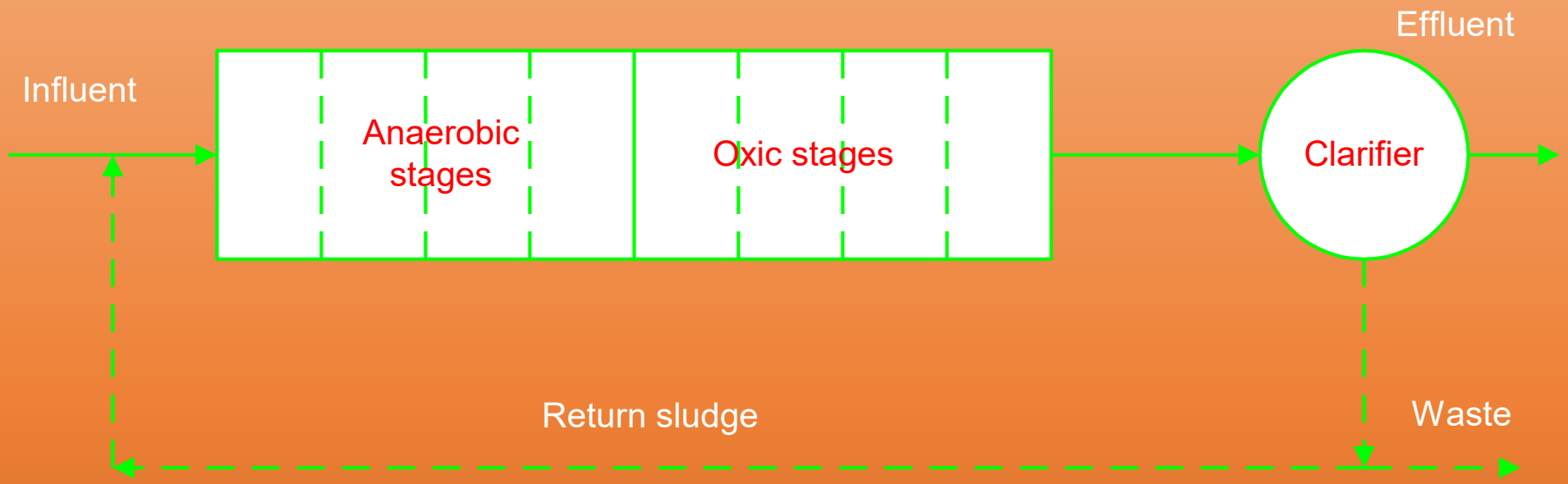


Fig. 12a. Biological phosphorus removal (mainstream process)

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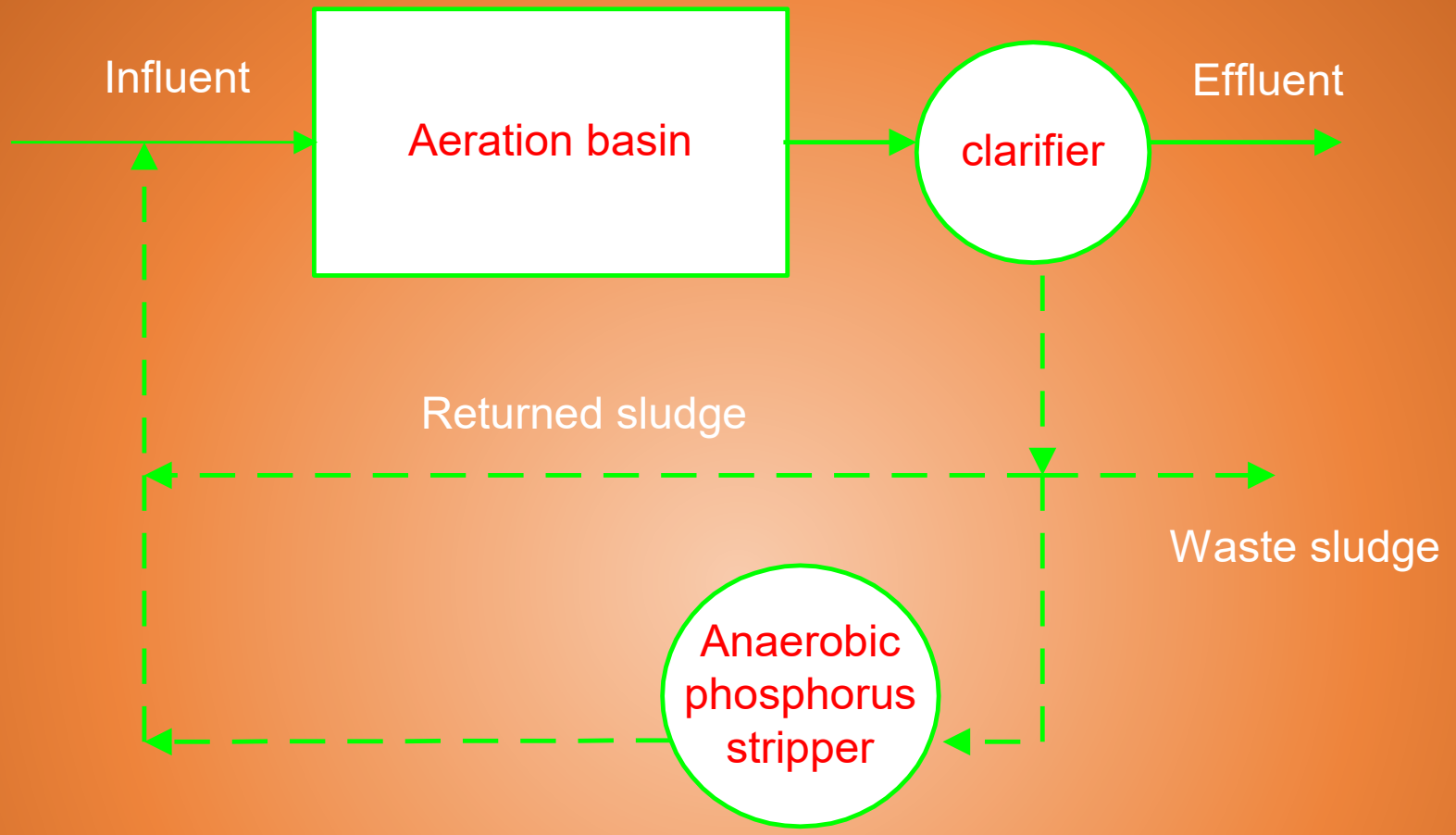


Fig. 12a. Biological phosphorus removal (sidestream process)

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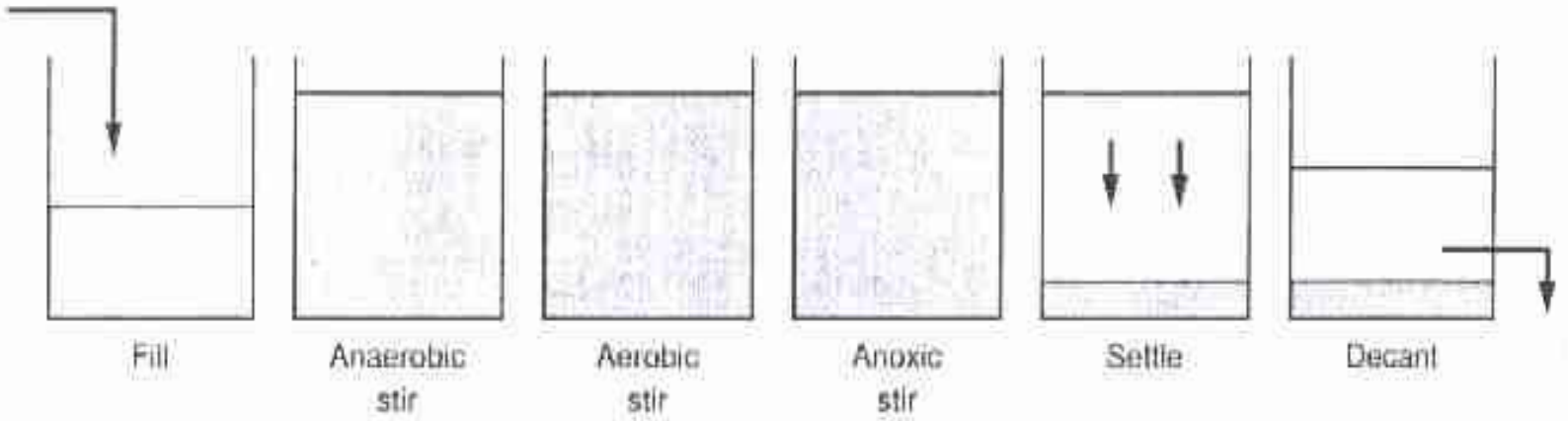


Fig. 13. Sequencing batch reactor operation for carbon, nitrogen, and phosphorus removal

Removal of Phosphorus (Chemically)

- Commonly used chemicals are
  - alum, sodium aluminate, ferric chloride, ferric sulfate, lime, and etc.
- Factors affecting the choice of chemicals
  - Influent phosphorus level
  - Wastewater suspended solids
  - Alkalinity
  - Chemical cost
  - Reliability of chemical supply
  - Sludge handling facilities
  - Ultimate disposal method
  - Compatibility with other treatment processes

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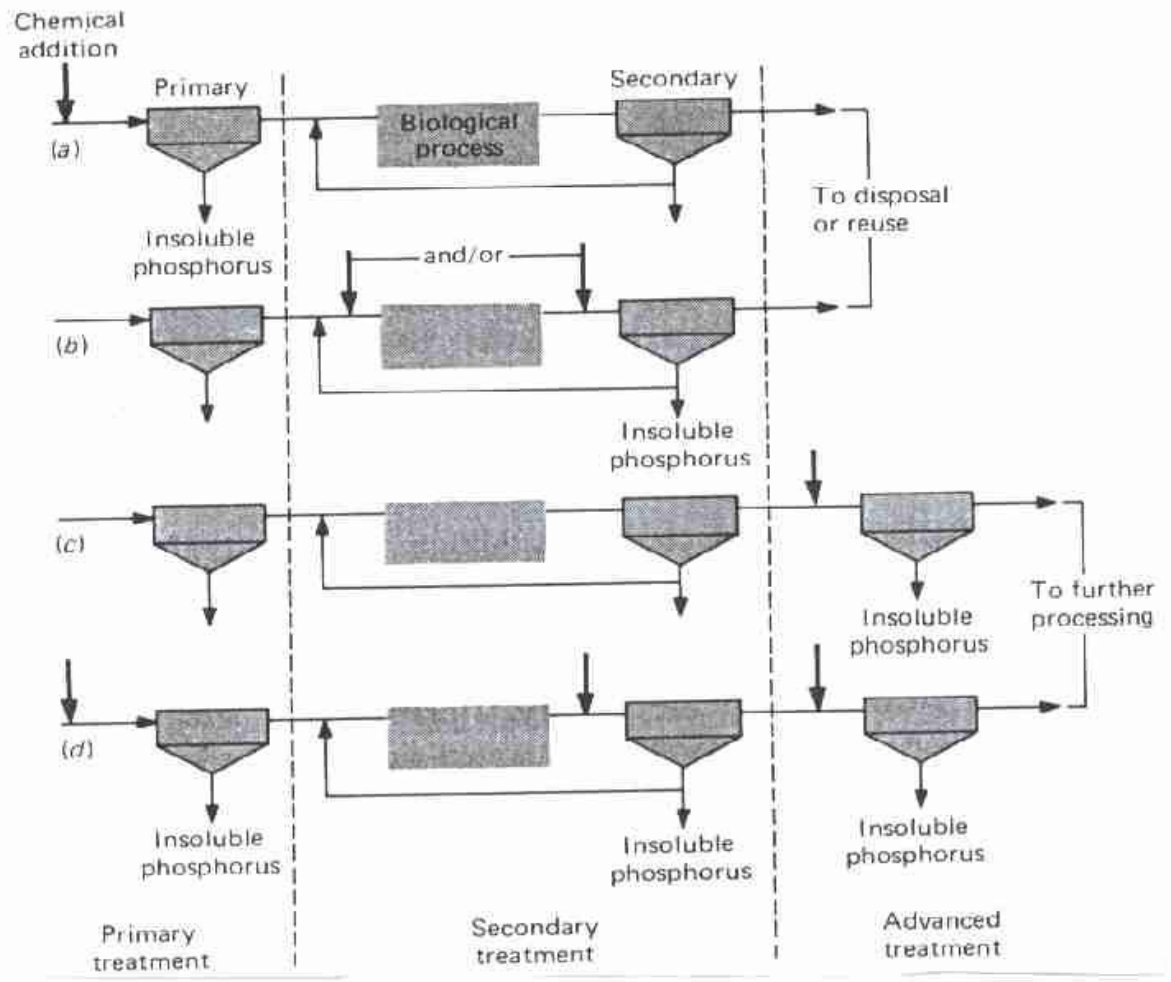


Fig. 15. Alternative points of chemical addition for phosphorus removal: (a) before primary sedimentation, (b) before and/or following biological treatment, (c) following secondary treatment, and (d) at several locations in a process

Table 13.0:

Effect of various treatment operations and processes on phosphorus removal[a]

Treatment operation or process	Removal of phosphorus entering system, %
Conventional treatment	
Primary	10–20
Activated-sludge	10–25
Trickling-filter	8–12
Rotating biological contactors	8–12
Biological phosphorus removal only	
Mainstream treatment	70–90
Sidestream treatment	70–90
Combined biological nitrogen and phosphorus removal	70–90
Chemical removal	
Precipitation with metal salt	70–90
Precipitation with lime	70–90
Physical removal	
Filtration	20–50
Reverse osmosis	90–100
Carbon adsorption	10–30

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(III) Removal of Toxic Compounds:

Special attention is given to priority pollutants & refractory organic compounds in recent years, due to:

- carcinogenic
- mutagenic
- teratogenic
- they are resistant to microbial degradation

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- Treatment methods
  - Biological
  - Chemical
    - chemical oxidation
    - coagulation, sedimentation, and filtration
  - Physical
    - carbon adsorption
    - air stripping

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- (1) Carbon Adsorption:

  It is an advanced wastewater treatment method used for the removal of refrectory organic compounds as well as residual amount of inorganic compounds

- Types of carbon contactors:
  - Upflow columns
  - Downflow columns
  - Fixed beds
  - Expanded beds

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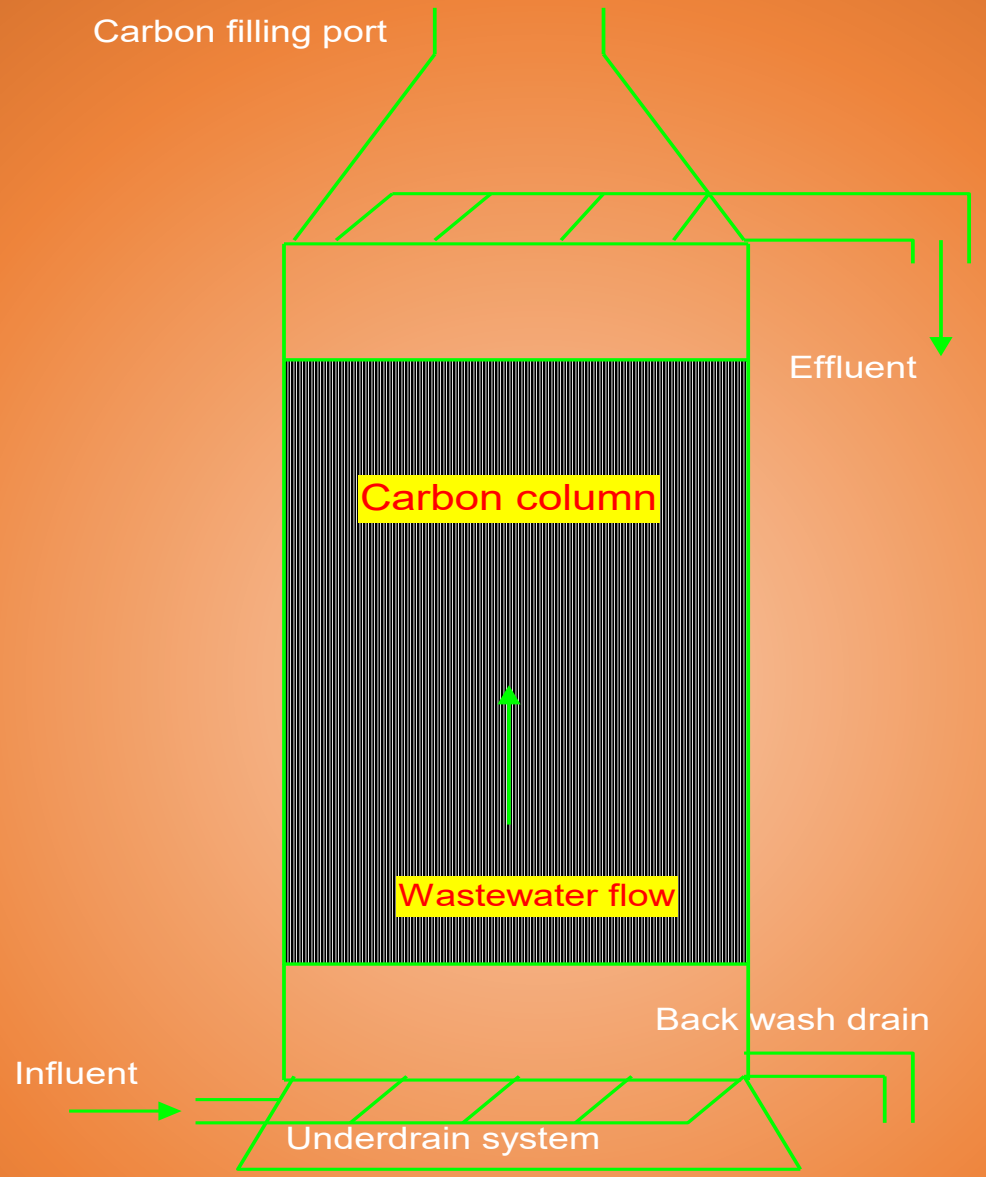


Fig. 17. Typical upflow countercurrent carbon column

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(2) Chemical Oxidation:

Chemical oxidation mainly done by

- chlorine
- chlorine dioxide, and
- ozone

<u>Basic chemical equation</u>:

$$\text{Oxidant} + \text{Compound} \rightarrow CO_2 + H_2O + \text{other products}$$

(IV) Removal of Dissolved Inorganic Compounds

- chemical precipitation
- ion exchange
- ultra-filtration
- reverse osmosis
- electrodialysis

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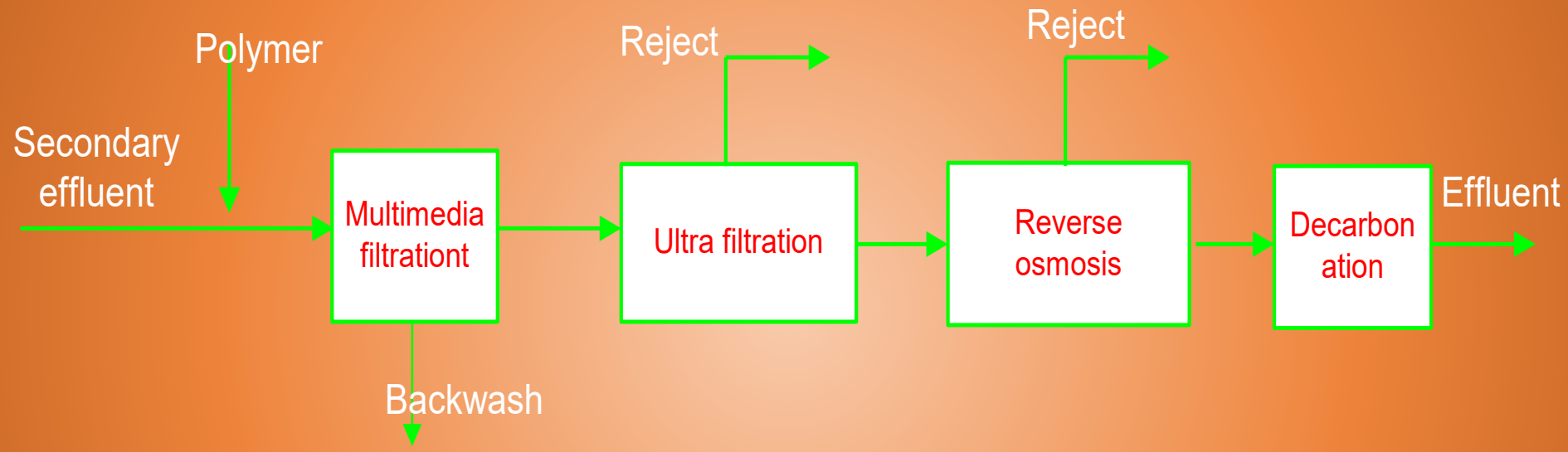


Fig. 19. Ultrafiltration and reverse osmosis for the removal of dissolved organics

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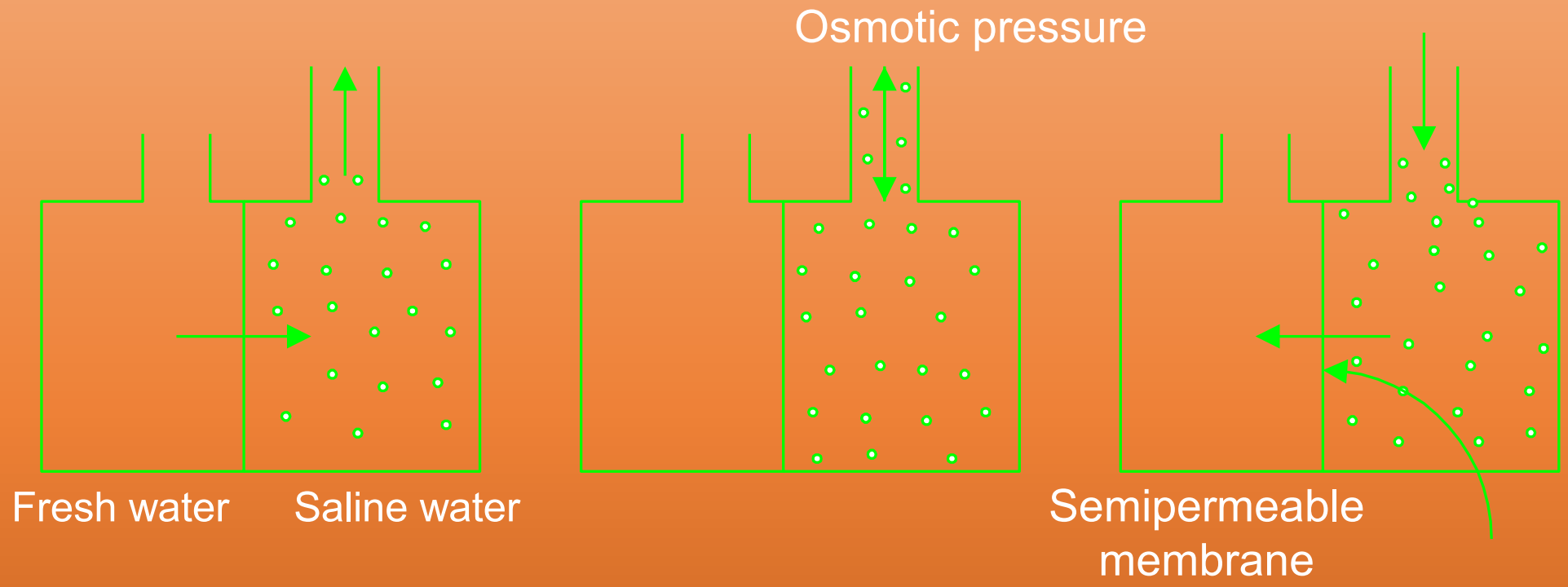


Fig. 20. Processes of reverse osmosis (a) direct osmosis, (b) osmotic equilibrium, (c) reverse osmosis

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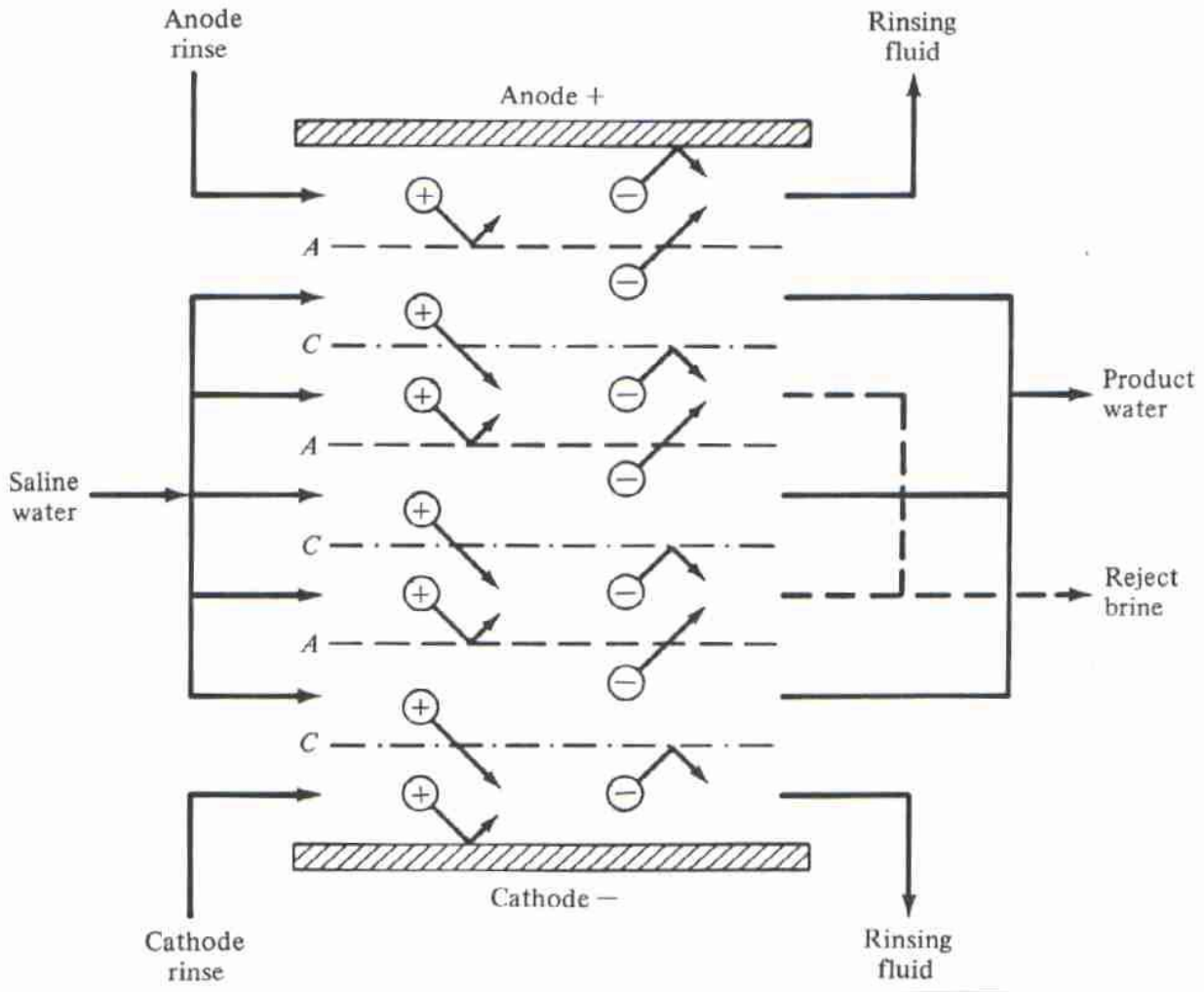


Fig. 24. Electrodialysis processes scheme for desalting water (membranes labeled A are permeable to anions only; those labeled C are permeable to cations only)

Tertiary Treatment of Wastewater in Saudi Arabia

- Large quantity of wastewater is being generated in kingdom of Saudi Arabia

- Quantities of wastewater generated
  - In 1994: Water Demand = 1.8 billion m^3
  - WW Generated = 1.0 billion m^3
  - WW Treated = 0.4 billion m^3
  - WW Recycled = 0.1 billion m^3
  - Water Demand in year 2000 = 2.8 billion m^3

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Table 21: Present and projected flow of wastewater, generated (m3/d) in three cities of kingdom

YEAR	DAMMAM	AL-KHOBAR	QATIF
2000	242,057	118,539	100,766
2005	291,324	137,419	120,735
2010	326,985	159,306	144,785
2015	375,794	187,100	173,627

Source Al-Elaiw, M. (1994).

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- Secondary treatment is practiced in Dammam, Khobar, Qatif and Khafji

- Tertiary treatment is practiced in Royal Commission of Jubail and Yanbu (RCJY)

- In Jubail 100% of tertiary treated wastewater is being reused

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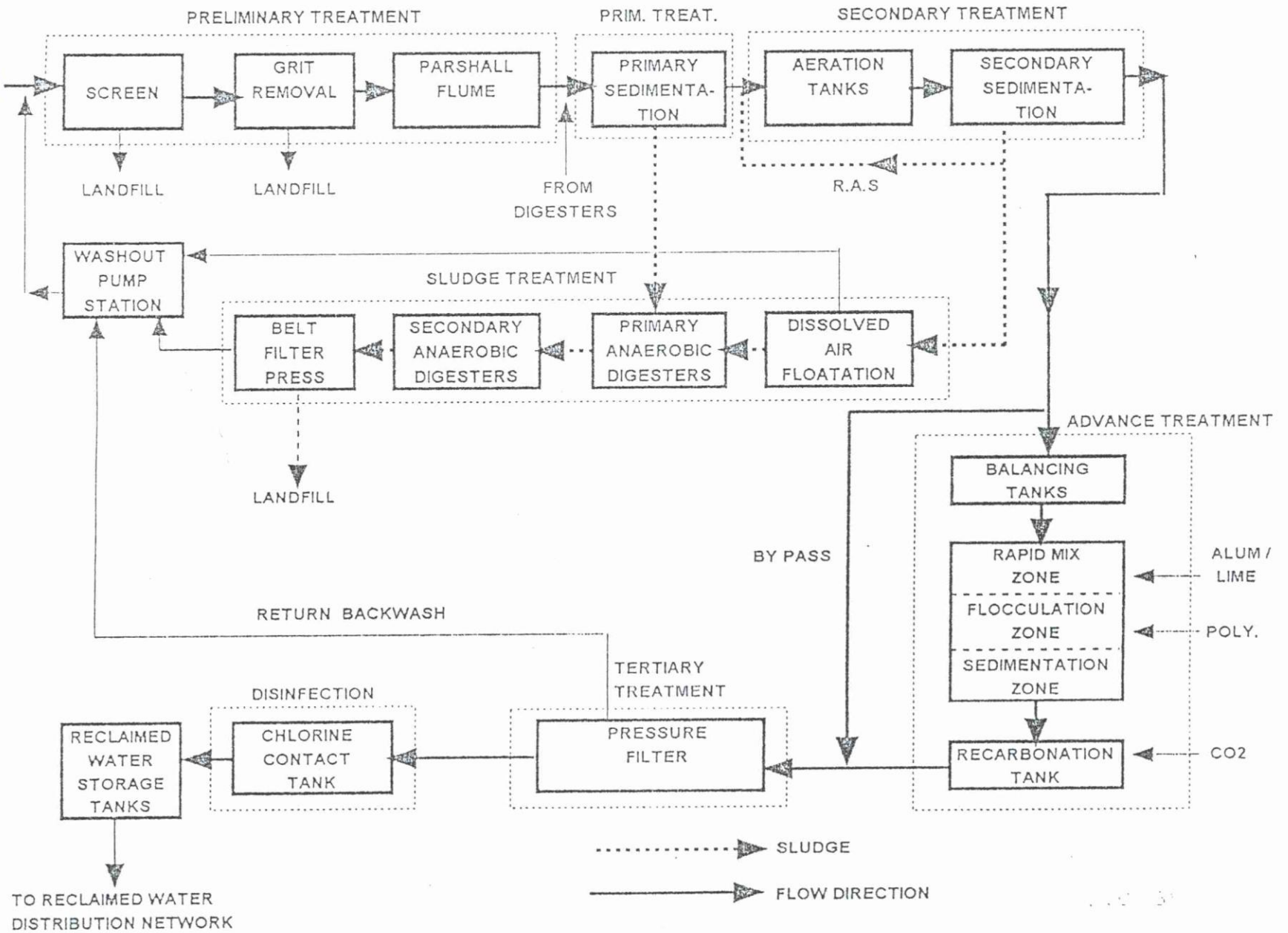


Fig. 25. Sewage water treatment plant flow diagram at Madinat Al-Jubail Al-Sinaiyah

- Summary:
  - Growing demand and scarcity of water resources necessitate the need for the tertiary treatment of wastewater for reuse purposes
  - Tertiary treatment of wastewater mainly depends on the availability and practicality of technologies
  - Selection of the processes depends on the requirement
  - Residual contaminants to be removed during tertiary treatment are suspended solids, nutrients, toxic compounds, and dissolved inorganics

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Tertiary Treatment

When the intended receiving water is very vulnerable to the effects of pollution, secondary effluent may be treated further by several tertiary processes.

Effluent Polishing

- For the removal of additional suspended solids and BOD from secondary effluent, effluent polishing is an effective treatment.
- It is most often accomplished using granular media filters, much like the filters used to purify drinking water.
- Polishing filters are usually built as prefabricated units, with tanks placed directly above the filters for storing backwash water.
- Effluent polishing of wastewater may also be achieved using microstrainers of the type used in treating municipal water supplies.

Tertiary Treatment

Removal of plant nutrients

- When treatment standards require the removal of plant nutrients from the sewage, it is often done as a tertiary step.
- Phosphorus in wastewater is usually present in the form of organic compounds and phosphates that can easily be removed by chemical precipitation.
- This process, however, increases the volume and weight of sludge.
- Nitrogen, another important plant nutrient, is present in sewage in the form of ammonia and nitrates.
- Ammonia is toxic to fish, and it also exerts an oxygen demand in receiving waters as it is converted to nitrates.
- Nitrates, like phosphates, promote the growth of algae and the eutrophication of lakes.
- Nitrates, like phosphates, promote the growth of algae and the eutrophication of lakes. A method called nitrification-denitrification can be used to remove the nitrates.

Tertiary Treatment

Removal of plant nutrients

- It is a two-step biological process in which ammonia nitrogen is first converted into nitrates by microorganisms.
- The nitrates are further metabolized by another species of bacteria, forming nitrogen gas that escapes into the air.
- This process requires the construction of more aeration and settling tanks and significantly increases the cost of treatment.
- A physicochemical process called ammonia stripping may be used to remove ammonia from sewage. Chemicals are added to convert ammonium ions into ammonia gas. The sewage is then cascaded down through a tower, allowing the gas to come out of solution and escape into the air. Stripping is less expensive than nitrification-denitrification, but it does not work very efficiently in cold weather.

Tertiary Treatment

Land treatment

- In some locations, secondary effluent can be applied directly to the ground and a polished effluent obtained by natural processes as the wastewater flows over vegetation and percolates through the soil.
- There are three types of land treatment: slow-rate, rapid infiltration, and overland flow.
- In the slow-rate, or irrigation, method, effluent is applied onto the land by ridge-and-furrow spreading (in ditches) or by sprinkler systems.
- Most of the water and nutrients are absorbed by the roots of growing vegetation.
- In the rapid infiltration method, the wastewater is stored in large ponds called recharge basins. Most of it percolates to the groundwater, and very little is absorbed by vegetation.
- For this method to work, soils must be highly permeable. In overland flow, wastewater is sprayed onto an inclined vegetated terrace and slowly flows to a collection ditch.

Tertiary Treatment

Land treatment

- Purification is achieved by physical, chemical, and biological processes, and the collected water is usually discharged into a nearby stream.
- Land treatment of sewage can provide moisture and nutrients for the growth of vegetation, such as corn or grain for animal feed.
- It also can recharge, or replenish, groundwater aquifers. Land treatment, in effect, allows sewage to be recycled for beneficial use.
- Large land areas are required, however, and the feasibility of this kind of treatment may be limited further by soil texture and climate.

Tertiary Treatment

Clustered wastewater treatment systems

- In certain instances when it is not feasible to connect residences or units to public sewer systems, communities may opt for a clustered wastewater treatment system.
- Such facilities are smaller versions of centralized treatment plants and serve only a limited number of connections.
- The technologies used for clustered wastewater treatment may be the same as those used for centralized systems or for individual on-site systems, depending upon the specific applications and degree of treatment required.
- Upon treatment, effluent from clustered wastewater systems can be discharged via surface or subsurface disposal methods.

Tertiary Treatment

- In sparsely populated suburban or rural areas, it is usually not economical to build sewage collection systems and a centrally located treatment plant.
- Instead, a separate treatment and disposal system is provided for each home.
- On-site systems provide effective, low-cost, long-term solutions for wastewater disposal as long as they are properly designed, installed, and maintained.
- In the United States, about one-third of private homes make use of an on-site subsurface disposal system.
- The most common type of on-site system includes a buried, watertight septic tank and a subsurface absorption field (also called a drain field or leaching field).

Tertiary Treatment

Where unfavourable site or soil conditions prohibit the use of both absorption fields and seepage pits, mound systems may be utilized for on-site sewage disposal. A mound is an absorption field built above the natural ground surface in order to provide suitable material for percolation and to separate the drain field from the water table. Septic tank effluent is intermittently pumped from a chamber and applied to the mound. Other alternative on-site disposal methods include use of intermittent sand filters or of small, prefabricated aerobic treatment units. Disinfection (usually by chlorination) of the effluent from these systems is required when the effluent is discharged into a nearby stream.

Waste water reuse

- Wastewater can be a valuable resource in cities or towns where population is growing and water supplies are limited.
- In addition to easing the strain on limited freshwater supplies, the reuse of wastewater can improve the quality of streams and lakes by reducing the effluent discharges that they receive.
- Wastewater may be reclaimed and reused for crop and landscape irrigation, groundwater recharge, or recreational purposes.
- Reclamation for drinking is technically possible, but this reuse faces significant public resistance.
- There are two types of wastewater reuse: direct and indirect.
- In direct reuse, treated wastewater is piped into some type of water system without first being diluted in a natural stream or lake or in groundwater.

Waste water reuse

- One example is the irrigation of a golf course with effluent from a municipal wastewater treatment plant.
- Indirect reuse involves the mixing of reclaimed wastewater with another body of water before reuse.
- In effect, any community that uses a surface water supply downstream from the treatment plant discharge pipe of another community is indirectly reusing wastewater.
- Indirect reuse is also accomplished by discharging reclaimed wastewater into a groundwater aquifer and later withdrawing the water for use.
- Discharge into an aquifer (called artificial recharge) is done by either deep-well injection or shallow surface spreading.

Waste water reuse

- Quality and treatment requirements for reclaimed wastewater become more stringent as the chances for direct human contact and ingestion increase.
- The impurities that must be removed depend on the intended use of the water.
- For example, removal of phosphates or nitrates is not necessary if the intended use is landscape irrigation.
- If direct reuse as a potable supply is intended, tertiary treatment with multiple barriers against contaminants is required.
- This may include secondary treatment followed by granular media filtration, ultraviolet radiation, granular activated carbon adsorption, reverse osmosis, air stripping, ozonation, and chlorination.

Waste water reuse

- The use of gray-water recycling systems in new commercial buildings offers a method of saving water and reducing total sewage volumes.
- These systems filter and chlorinate drainage from tubs and sinks and reuse the water for nonpotable purposes (e.g., flushing toilets and urinals).
- Recycled water can be marked with a blue dye to ensure that it is not used for potable purposes.

Sludge treatment and disposal

- The residue that accumulates in sewage treatment plants is called sludge (or biosolids).
- Sewage sludge is the solid, semisolid, or slurry residual material that is produced as a by-product of wastewater treatment processes.
- This residue is commonly classified as primary and secondary sludge. Primary sludge is generated from chemical precipitation, sedimentation, and other primary processes, whereas secondary sludge is the activated waste biomass resulting from biological treatments.
- Some sewage plants also receive septage or septic tank solids from household on-site wastewater treatment systems.
- Quite often the sludges are combined together for further treatment and disposal.
- Two basic goals of treating sludge before final disposal are to reduce its volume and to stabilize the organic materials. Stabilized sludge does not have an offensive odour and can be handled without causing a nuisance or health hazard. Smaller sludge volume reduces the costs of pumping and storage.

Sludge treatment and disposal

Treatment methods

Treatment of sewage sludge may include a combination of thickening, digestion, and dewatering processes.

Thickening

Thickening is usually the first step in sludge treatment because it is impractical to handle thin sludge, a slurry of solids suspended in water. Thickening is usually accomplished in a tank called a gravity thickener. A thickener can reduce the total volume of sludge to less than half the original volume. An alternative to gravity thickening is dissolved-air flotation. In this method, air bubbles carry the solids to the surface, where a layer of thickened sludge forms.

Sludge treatment and disposal

Digestion

Sludge digestion is a biological process in which organic solids are decomposed into stable substances. Digestion reduces the total mass of solids, destroys pathogens, and makes it easier to dewater or dry the sludge. Digested sludge is inoffensive, having the appearance and characteristics of a rich potting soil.

Most large sewage treatment plants use a two-stage digestion system in which organics are metabolized by bacteria anaerobically (in the absence of oxygen). In the first stage, the sludge, thickened to a dry solids (DS) content of about 5 percent, is heated and mixed in a closed tank for several days. Acid-forming bacteria hydrolyze large molecules such as proteins and lipids, breaking them into smaller water-soluble molecules, and then ferment those smaller molecules into various fatty acids. The sludge then flows into a second tank, where the dissolved matter is converted by other bacteria into biogas, a mixture of carbon dioxide and methane. Methane is combustible and is used as a fuel to heat the first digestion tank as well as to generate electricity for the plant.

Sludge treatment and disposal

Anaerobic digestion is very sensitive to temperature, acidity, and other factors. It requires careful monitoring and control. In some cases, the sludge is inoculated with extra hydrolytic enzymes at the beginning of the first digestion stage in order to supplement the action of the bacteria. It has been found that this enzymatic treatment can destroy more unwanted pathogens in the sludge and also can result in the generation of more biogas in the second stage of digestion.Another enhancement of the traditional two-stage anaerobic digestion process is thermal hydrolysis, or the breaking down of the large molecules by heat. This is done in a separate step before digestion. In a typical case, the process begins with a sludge that has been dewatered to a DS content of some 15 percent.

Sludge treatment and disposal

The sludge is mixed with steam in a pulper, and this hot homogenized mixture is fed to a reactor, where it is held under pressure at approximately 165 °C (about 330 °F) for about 30 minutes. At that point, with the hydrolytic reactions complete, some of the steam is bled off (to be fed to the pulper), and the sludge, still under some pressure, is released suddenly into a “flash tank,” where the sudded drop in pressure bursts the cell walls of much of the solid matter. The hydrolyzed sludge is cooled, diluted slightly with water, and then sent directly to the second stage of anaerobic digestion.

Sludge treatment and disposal

Sludge digestion may also take place aerobically—that is, in the presence of oxygen. The sludge is vigorously aerated in an open tank for about 20 days. Methane gas is not formed in this process. Although aerobic systems are easier to operate than anaerobic systems, they usually cost more to operate because of the power needed for aeration. Aerobic digestion is often combined with small extended aeration or contact stabilization systems.

Aerobic and conventional anaerobic digestion convert about half of the organic sludge solids to liquids and gases. Thermal hydrolysis followed by anaerobic digestion can convert some 60 to 70 percent of the solid matter to liquids and gases. Not only is the volume of solids produced smaller than in conventional digestion, but the greater production of biogas can make some wastewater treatment plants self-sufficient in energy.

Sludge treatment and disposal

Dewatering

Digested sewage sludge is usually dewatered before disposal. Dewatered sludge still contains a significant amount of water—often as much as 70 percent—but, even with that moisture content, sludge no longer behaves as a liquid and can be handled as a solid material. Sludge-drying beds provide the simplest method of dewatering. A digested sludge slurry is spread on an open bed of sand and allowed to remain until dry. Drying takes place by a combination of evaporation and gravity drainage through the sand. A piping network built under the sand collects the water, which is pumped back to the head of the plant. After about six weeks of drying, the sludge cake, as it is called, may have a solids content of about 40 percent. It can then be removed from the sand with a pitchfork or a front-end loader. In order to reduce drying time in wet or cold weather, a glass enclosure may be built over the sand beds.

Sludge treatment and disposal

Dewatering

Since a good deal of land area is needed for drying beds, this method of dewatering is commonly used in rural or suburban towns rather than in densely populated cities.

Alternatives to sludge-drying beds include the rotary drum vacuum filter, the centrifuge, and the belt filter press. These mechanical systems require less space than do sludge-drying beds, and they offer a greater degree of operational control. However, they usually have to be preceded by a step called sludge conditioning, in which chemicals are added to the liquid sludge to coagulate solids and improve drainability.

Sludge treatment and disposal

Disposal

The final destination of treated sewage sludge usually is the land. Dewatered sludge can be buried underground in a sanitary landfill. It also may be spread on agricultural land in order to make use of its value as a soil conditioner and fertilizer. Since sludge may contain toxic industrial chemicals, it is not spread on land where crops are grown for human consumption.

Where a suitable site for land disposal is not available, as in urban areas, sludge may be incinerated. Incineration completely evaporates the moisture and converts the organic solids into inert ash. The ash must be disposed of, but the reduced volume makes disposal more economical. Air pollution control is a very important consideration when sewage sludge is incinerated. Appropriate air-cleaning devices such as scrubbers and filters must be used.

Dumping sludge in the ocean, once an economical disposal method for many coastal communities, is no longer considered a viable option. It is now prohibited in the United States and many other coastal countries.

Circular economy on waste water treatment

The circular economy (CE) is a concept in which goods, materials (including raw materials), and waste are viewed as secondary raw materials that can be recycled for processing and re-use. This sets it apart from a linear economy based on the 'take-make-use-dispose' paradigm, in which waste is often the product's last step cycle of life.



https://www.netsolwater.com/reusing-waste-water-to-support-circular-economy.php?blog=909

Circular economy in waste water treatment

CE is a concept that encourages sustainable material and energy management by minimising waste, the amount of waste produced and its potential for re-use as a secondary material.The main reasons for implementing a circular economy are limited raw material availability, the economy's reliance on raw material imports (high prices, market volatility, and uncertain political situations in certain countries), and the economy's decreasing competitiveness in global economies. The reuse of treated **wastewater** from **WWTPs** for agriculture and land irrigation, industrial purposes, toilet flushing, and groundwater replenishment is a key component of the current strategy aimed at releasing freshwater for domestic use, improving the effluent quality of WWTPs, and, as a result, improving the quality of river waters used for drinking water abstraction.

Circular economy goal

The goal of circular economy is to optimise circle system functions by reducing resources exiting the system. Moving toward greater circular economy can assist in meeting the resource efficiency goals set out in the Strategy for smart, inclusive, and sustainable growth.

For optimal sustainable development, smart cities concept is a pragmatic and balanced blend of social, economic, ecological, and other key fields. The purpose of ecological cities around the world is to "improve citizens' and society's well-being through integrated urban planning and management that fully utilises the benefits of ecological systems while protecting and nurturing these assets for future generations."

Solid waste sources

Source	Typical waste generators	Types of solid wastes
Residential	Single and multifamily dwellings	Food wastes, paper, cardboard, plastics, textiles, leather, yard wastes, wood, glass, metals, ashes, special wastes (e.g., bulky items, consumer electronics, white goods, batteries, oil, tires), and household hazardous wastes.).
Industrial	Light and heavy manufacturing, fabrication, construction sites, power and chemical plants.	Housekeeping wastes, packaging, food wastes, construction and demolition materials, hazardous wastes, ashes, special wastes.
Commercial	Stores, hotels, restaurants, markets, office buildings, etc.	Paper, cardboard, plastics, wood, food wastes, glass, metals, special wastes, hazardous wastes.



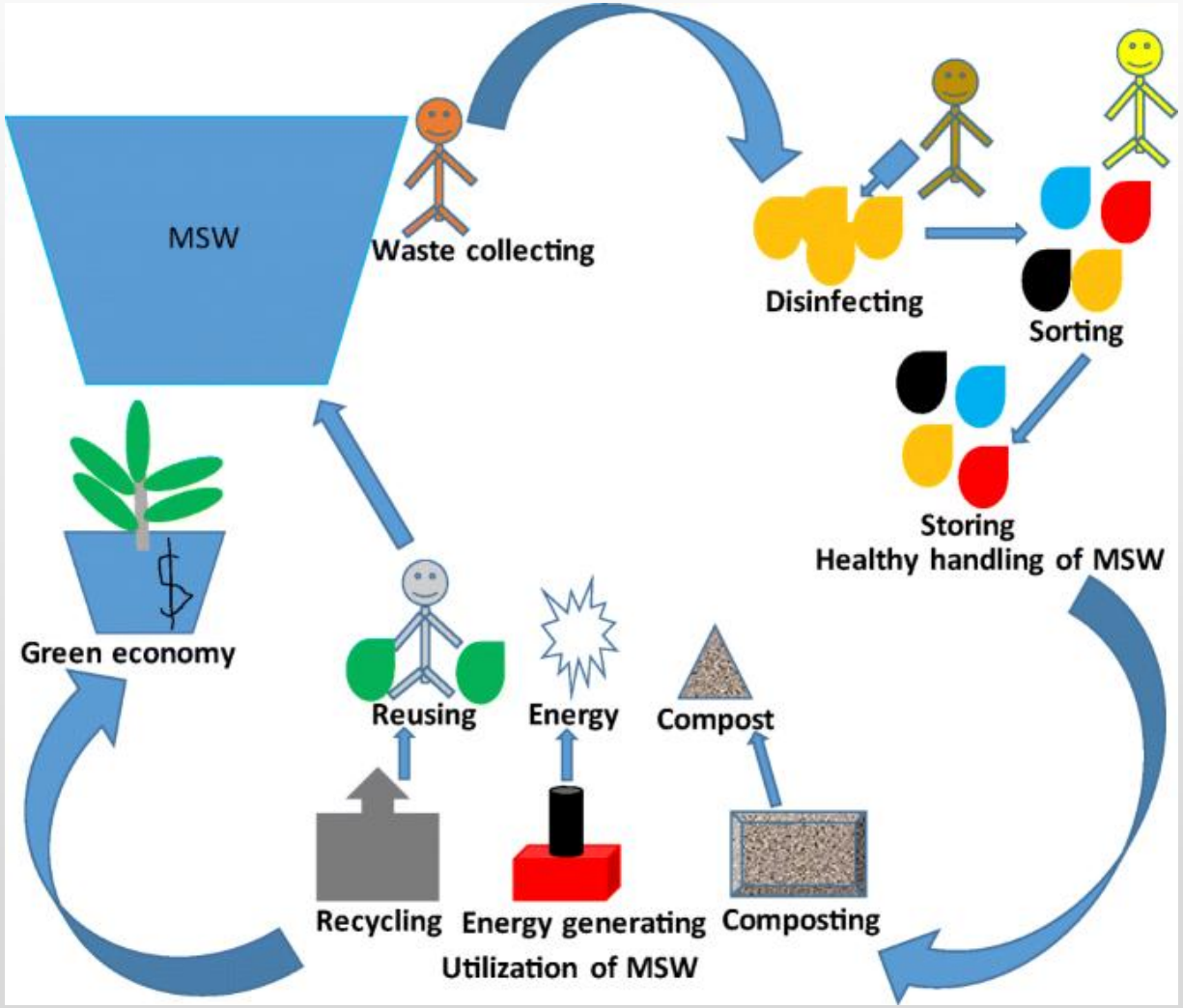
Solid waste sources

Source	Typical waste generators	Types of solid wastes
Institutional	Schools, hospitals, prisons, government centers.	Same as commercial.
Construction and demolition	New construction sites, road repair, renovation sites, demolition of buildings	Wood, steel, concrete, dirt, etc.
Municipal services	Street cleaning, landscaping, parks, beaches, other recreational areas, water and wastewater treatment plants.	Street sweepings; landscape and tree trimmings; general wastes from parks, beaches, and other recreational areas; sludge.
Process (manufacturing, etc.)	Heavy and light manufacturing, refineries, chemical plants, power plants, mineral extraction and processing.	Industrial process wastes, scrap materials, off-specification products, slay, tailings.
Agriculture	Crops, orchards, vineyards, dairies, feedlots, farms.	Spoiled food wastes, agricultural wastes, hazardous wastes (e.g., pesticides).



Integrated solid waste management

➢Integrated solid waste management (ISWM) refers to the strategic approach to sustainable management of solid wastes covering all sources and all aspects such as waste generation, segregation, transfer, sorting, treatment, recovery and disposal in an integrated manner, with an emphasis on maximizing resource use efficiency.



https://link.springer.com/article/10.1007/s11356-021-13914-6

Integrated solid waste management

- Include collective management of all types of wastes and implementation of the 3R (Reduce, Reuse and Recycle) policies and strategies.
- An efficient ISWM system considers how to reduce, reuse, recycle, and manage waste to protect human health and the natural environment.
- It involves evaluating local conditions and needs. Then choosing, mixing and applying the most suitable solid waste management activities according to the condition.



https://stock.adobe.com/in/search?k=reduce+reuse+recycle

The importance of ISWM as a waste management approach

➢With rapid population expansion and constant economic development, waste generation both in residential as well as commercial/industrial areas continues to grow rapidly, putting pressure on society's ability to process and dispose of this material.

➢Also, inappropriately managed solid waste streams can pose a significant risk to health and environmental concerns. Improper waste handling in conjunction with uncontrolled waste dumping can cause a broad range of problems, including polluting water, attracting rodents and insects, as well as increasing floods due to blockage in drains.



https://www.netsolwater.com/what-is-solid-waste-management.php?blog=3204

The importance of ISWM as a waste management approach

- Improper solid waste management can also increase greenhouse gas (GHG) emissions, thus contributing to climate change.
- Having a comprehensive waste management system for efficient waste collection, transportation, and systematic waste disposal—together with activities to reduce waste generation and increase waste recycling—can significantly reduce all these problems. While nothing new, an ISWM approach provides the opportunity to create a suitable combination of existing waste management practices to manage waste most efficiently.



https://www.zerozen.co.uk/disposable-vs-reusable-water-bottles-and-travel-cups/



Functional elements of integrated solid waste management

- The four components or functional elements of ISWM include source reduction, recycling and composting, waste transportation and landfilling.

- **Source Reduction**, also known as waste prevention, aims at reducing unnecessary waste generation.

- Source reduction strategies may include a variety of approaches, such as:
  - products that are designed for recycling, durable, sustainable goods and, where possible, in concentrated form.
  - reusable products, including reusable packaging, as reuse and increasingly becomes an important component of the circular economy.
  - refurbishing of goods to prolong product life, another important element of the circular economy model.
  - redesign of goods and utilize less or no packaging.
  - reduction of food spoilage and waste through better attention to food processing and storage

- Waste source reduction helps us to lessen waste handling, transportation, and disposal costs and eventually reduces methane generation.

Functional elements of integrated solid waste management

- **Recycling and Composting** are crucial phases in the entire ISWM process.
- Recycling includes the accumulation, sorting and recovering of recyclable and reusable materials, as well as the reprocessing of recyclables to produce new products.
- Composting, a component of organics recycling, involves the accumulation of organic waste and converting it into soil additives.
- Both recycling and composting wastes have a number of economic benefits such as they create job opportunities in addition to diverting material from the waste stream to generate cost-effective sources of material for further use.
- Both recycling and composting also significantly contribute to the reduction of greenhouse gas emissions.

Functional elements of integrated solid waste management

- **Waste Transportation** is another waste management activity that must be integrated systematically with other waste management activities to ensure smooth and efficient waste management. Typically this includes the collection of waste from curbside and businesses, as well as from transfer stations where waste may be concentrated and reloaded onto other vehicles for delivery to the landfill.
- **Waste Disposal,** in particular through the use of landfills and combustion, are the activities undertaken to manage waste materials that are not recycled. The most common way of managing these wastes is through landfills, which must be properly designed, well-constructed and systematically managed.

Hazardous waste

Meaning of Hazardous Waste:

According to USEPA, hazardous waste is defined as any solid, liquid or gaseous substance which, by reasons of any physical, chemical, reactive, toxic or infectious characteristics, causes danger to health or environment when improperly treated, stored, transported or disposed off.

Sources of Hazardous Waste:

Sources of hazardous wastes include industries, petroleum refineries, transportation, coke-ovens, blast furnace sludges, scrubbing sludges and biological waste generating laboratories, thermal and nuclear power plants and nuclear reactors etc.

Classification of hazardous wastes

➢Wastes are three categories such as toxic, inflammable and radioactive etc. Such a classification system helps the fire service to properly deal with the hazardous wastes under emergencies.

➢Many countries (e.g., UK, France, Germany and Netherlands) follow a Supplementary classification system for hazardous substances along with the general classification system. However, in Japan, the four types of wastes viz. sludges, slags, acidic waste and alkaline wastes, which contain any of the 9 materials (e.g., Cd, As, Hg, Pb, Cr (VI), CN, Se, PCBs and organic phosphates) beyond permissible limits, are considered to be hazardous.

➢Overall, toxicity, corrosivity, persistence, inflammability, quantity of waste involved, the extent of hazard to the environment and ultimate effects on living organisms.

As regards human toxicity, a waste is considered to be hazardous if:

(a) It has an oral LD 50 toxicity equal to or greater than 50 mg/kg (i.e., LD 50 is the lethal dose of the toxic waste at which 50% of the experimental animals die as a result of oral ingestion).

(b) It has an inhalation LC 50 toxicity of 2 mg/kg (LC 50 is the lethal ambient concentration of the toxic material in mg/L of air causing 50% mortality to test rats during 4 hour inhalation).

Test for hazardous waste

- Ignitability that is, whether the waste causes or enhances fire.
- Corrosivity that is, whether the waste destroys the tissues or metal.
- Toxicity that is, whether the substance pollutes water supplies and threatens the health.
- Reactivity that is, whether the substance reacts violently or causes explosion.
- As far as toxicity to the ground water is concerned, the 14 contaminants viz., 8 metals (Pb, Cd, Hg, As, Cr, Ba, Se, Ag), 4 pesticides (Lindane, endrin, toxaphene, methoxychlor) and 2 herbicides (2, 4-D and 2, 4, 5-T) are considered to be hazardous.

Hazardous waste management

The main objective of hazardous waste management plan is to eliminate or reduce the hazardous waste through process changes or resource recovery. The waste must be accounted for from its origin to ultimate disposal.

The major components of a hazardous waste management plan are as follows:

- **Inventory:** All the industries producing more than 100 kg of hazardous wastes per month should be registered. An inventory of such waste generating agencies and the quantity of waste produced per month should be prepared.
- **Storage:** The hazardous waste generating plants should equip themselves with special on-site tanks, chemically resistant containers or basins for storage of waste.
- **Transport:** The wastes must be transported by suitable tanker trucks, flat-bed trucks or rail cars of the approved disposal sites.

https://www.youtube.com/watch?v=7i0QMnz4ExY

https://www.youtube.com/watch?v=D8jfLf_bODs

https://youtu.be/D8jfLf_bODs?si=AP6z7iXVoPRZ2eL1

Hazardous waste management

Spillage:

➢A well publicized emergency plan should be prepared to deal with unexpected spillage or accidental release of contaminants during transport to prevent environmental damage and public health hazard.

Disposal:

➢The wastes should be transported to a physical and chemical treatment plants for processing or concentration or hauled directly to an approved hazardous waste treatment facility for final disposal.

Waste Exchange:

➢When on-site recycling or recovery of waste is not feasible, the possibility of waste exchange should be explored.

➢Transferring wastes from large industries to smaller ones which are able to reuse low purity oils, acids, alkalies, solvents or recover valuable metals and other materials from concentrated wastes, can also be considered.

Effect on human health

- Emergency response efforts must consider the health and ecological hazards of a hazardous substance release.
- These hazards impact emergency responders and effected communities. In some cases, hazardous substances may irritate the skin or eyes, make it difficult to breathe, cause headaches and nausea, or result in other types of illness.
- Some hazardous substances can cause far more severe health effects, including:
  - behavioral abnormalities,
  - cancer
  - physiological malfunctions (e.g., reproductive impairment, kidney failure, etc.),
  - physical deformations
  - birth defects.

E-waste sources and management

- Today's Electronic Gadgets, Tomorrow's E- Waste!
- Its no secret that E-Waste is increasing at an alarming rate with each passing day.
- All items of electrical and electronic equipment and its parts that have been discarded by the user as waste without the purpose of re-use or re-cycle is called Electronic Waste.
- Stored used electronics are also considered E-Waste.
- Improper processing of E-Waste can lead to dangerous human health effects and environmental pollution.



https://www.indiafilings.com/learn/e-waste-management/

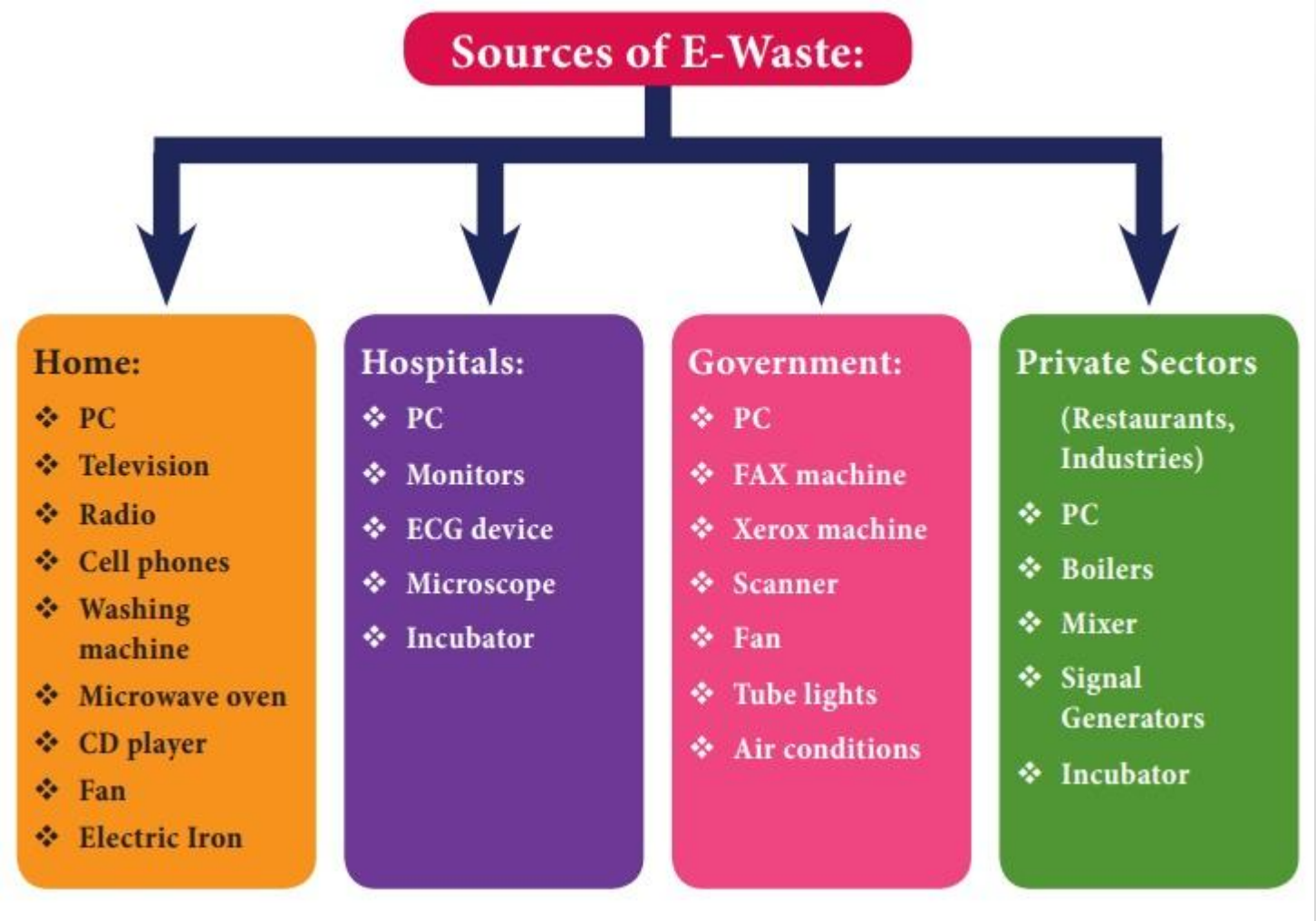
Sources of E-wastes

- Common things of electrical and electronic waste are:

Large household appliances like refrigerators/freezers, washing machines, dishwashers, televisions.

- Small household appliances which include toasters, coffee makers, irons, hairdryers.
- Information Technology (IT) and Telecommunications equipment namely personal computers, telephones, mobile phones, laptops, printers, scanners, photocopiers etc.
- Lighting equipment such as fluorescent lamps.
- Electronic or Electrical tools i.e. handheld drills, saws, screwdrivers etc.
- Toys and sports equipment.
- Monitoring and control instruments.
- Automatic dispensers.

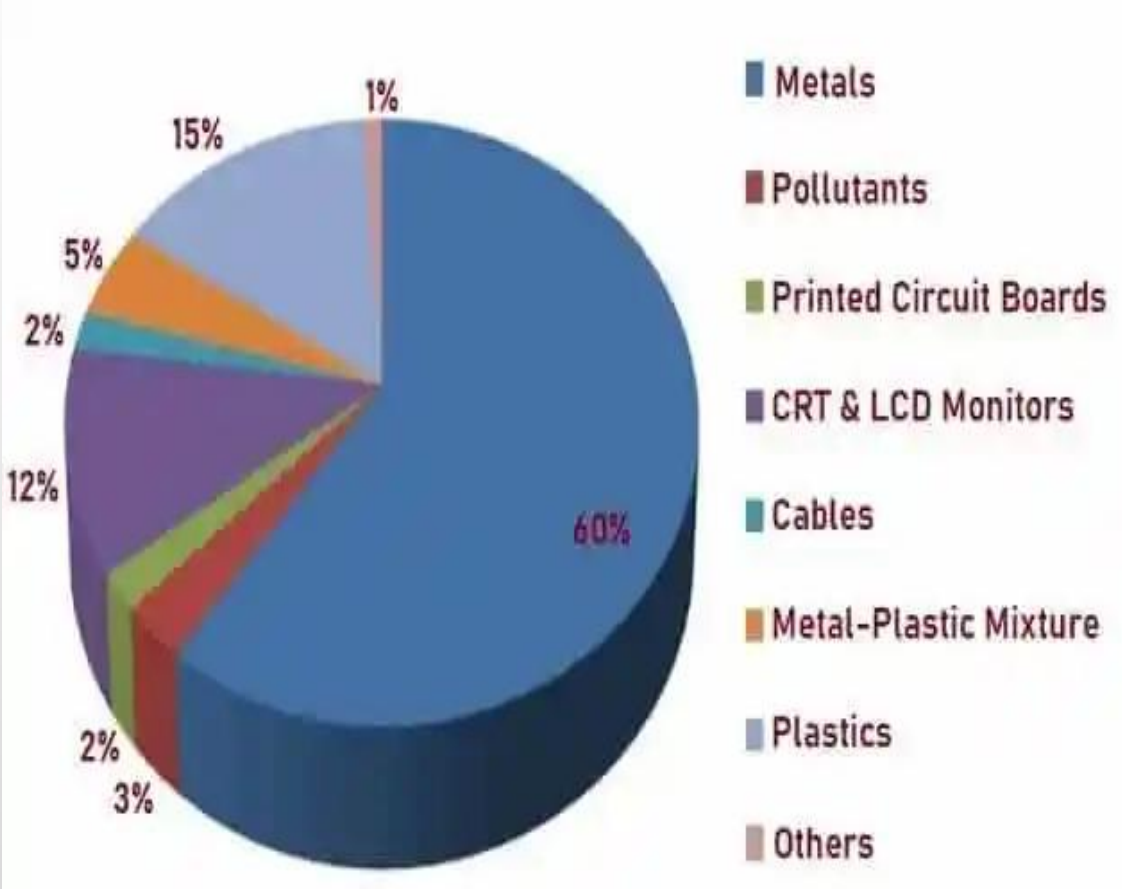
Sources of E-wastes



https://www.brainkart.com/article/e-Wastes_37147/

Causes of E-wastes

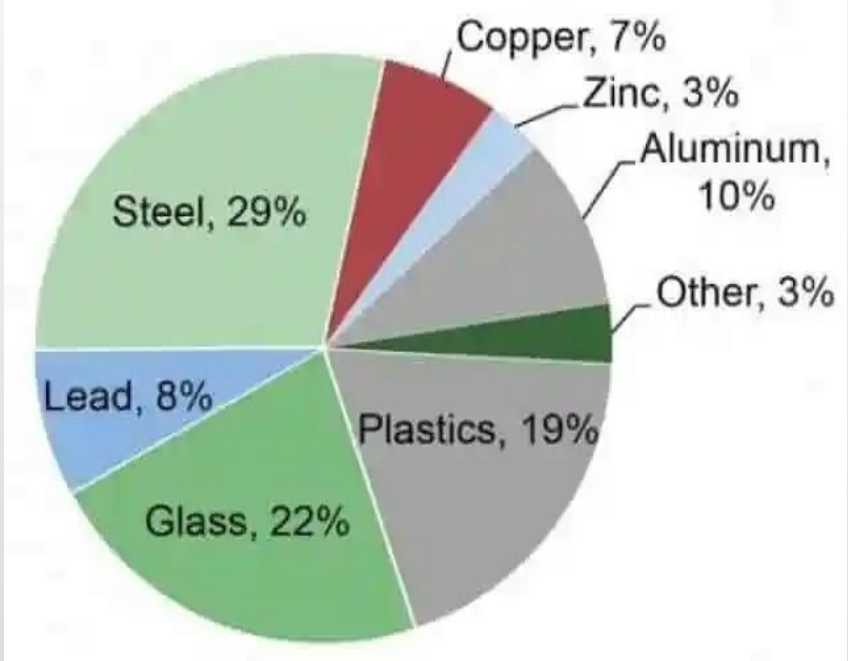
- Advancement in Technology.
- Changes in style fashion and status.
- End of their helpful life.
- Not taking precautions while handling them.



https://electricalfundablog.com/e-waste-electronic-waste-sources-composition/

Composition of E-waste

- Valuable metals like gold, platinum, silver and palladium.
- Useful metals like copper, aluminium, iron etc.
- Hazardous substances like radioactive isotopes and mercury.
- Toxic substances like Dioxins.
- Plastic like High Impact Polystyrene (HIPS), Acrylonitrile Butadiene Styrene (ABS), Polycarbonate (PC), Polyphenylene oxide (PPO) etc.
- Glass material like Cathode Ray Tube glass made up of SiO_2, CaO, Na.
- For instance, a mobile phone contains more than 40 elements, base metals such as Copper (Cu) and Tin (Sn), special metals such as Lithium (Li), Cobalt (Co), Indium (In) and Antimony (Sb) and precious metals such as Silver (Ag), Gold (Au), and Palladium (Pd).



https://electricalfundablog.com/e-waste-electronic-waste-sources-composition/

Effects of E-waste

Effects of E-Waste on Air: Most common result of E-waste on air is through air pollution. Burning of e waste can release hydrocarbons within the atmosphere that pollutes the air.

E-Waste Negatively Impacts Soil: E-waste can have a negative effect on the soil. As e-waste breaks down, it releases toxic heavy metals. Such heavy metals include lead, arsenic, and cadmium. When these toxins penetrate the soil, they influence the plants and trees. Thus, these toxins can enter the human food supply, which can lead to birth defects as well as a number of other health complications.

Effects of E-Waste on Water: Heavy metals like mercury, lithium, lead present in electronics (found in mobile phone and computer batteries), etc., when not disposed properly, these heavy metals penetrate from soil to groundwater which then run to the surface as streams or small ponds of water.

Collection, treatment and disposal system

Various methods of treatment and disposal system includes:

Land Filling– Disposal of Electronic Waste is mainly through land filling. Mostly, the discarded electronic goods finally end-up in landfill sites along with other municipal waste or are openly burnt releasing toxic and carcinogenic substances into the atmosphere.

Incineration– In this complete combustion process, the waste material is burned in specially designed incinerators at a high temperature (900-1000^{0}C). It reduces waste volume and some environmentally hazardous organic substances are converted into less hazardous compounds.



https://cpreecenvis.nic.in/Database/Payconsumerstotakebacke-waste_3318.aspx?format=Print



https://www.eurekalert.org/multimedia/751333

Collection, treatment and disposal system

➢**Recycling of E-Waste** – Recycling involves dismantling, processing and end processing. Comparatively, the value of recycling from the element could be much higher in comparison to other treatments.

➢ **Re-use**– It includes direct second hand use or use after slight modifications to the original functioning equipment like Inkjet cartridge is used after refilling. Old working computers can be donated to schools or organization working in the field of education. Computers beyond repairs can be returned back to the manufacturers. This can considerably reduce the volume of E-Waste generation converted into less hazardous compounds.



https://www.indiamart.com/proddetail/e-waste-dismantling-service-22410198988.html



https://thewestsider.com.au/reduce-reuse-and-recycle-electronic-devices-to-protect-our-environment/

e-waste recycling

- E-waste recycling then refers to the reprocessing and re-use of these electronic wastes. It is simple. It is a process that seeks to recover material from electronic waste. This way, you can use them in new electronic products.
- Recycling of e-waste is a growing trend and was initiated to protect human and environmental health mainly due to the widespread environmental pollution impacts of e-waste.
- Even more, millions of electronics are in use daily. Then, when they reach the tail of their lifespan, they mostly waste away in landfills.
- So, you will agree that e-waste recycling is an activity that you must encourage at any level. If you're someone who has developed an interest in recycling e-waste, or you're simply trying to build interest, you might want to ask: what components can we recycle?

Glass

➢You can extract glass from CRTs (Cathode Ray Tubes) of computer monitors and televisions. But there's a little problem here. CRTS contains several hazardous substances, such as lead. And this is dangerous to both human health and the immediate environment. This makes it difficult to retrieve a glass from CRTs. However, there are certain steps you may take to ensure safer CRT recycling.

➢First, separate the CRT from the monitor or television. Then shred the CRT into small pieces. Remove the metals with over-band magnets. This helps you remove ferrous and even non-ferrous objects from that glass.

➢After this, use washing lines to clear phosphors and oxides from that glass. The last step is called glass sorting. This is where you separate non-leaded from leaded gas. You can then use the extract to make newer screens.

Mercury

➢Devices containing mercury may be sent to recycling facilities using specialized technology to eliminate mercury. The end product of this elimination includes metric instruments, dental amalgams, and fluorescent lighting.

Circuit Boards

➢There are accredited and specialized companies smelting and recovering resources like tin, gold, silver, copper, palladium, and valuable metals.

Hard disk

- When shredded and processed, you can recover aluminum ingots from hard disks. These are particularly useful for automobiles.
- **Toner and Ink Cartridges** Recyclers in various manufacturing industries that remanufacture them take these toners and ink cartridges for recycling. They then use retrieved plastic and metals as raw materials for other products.
- **Batteries**You can take your scrap batteries to specialist recyclers to recover cadmium, steel, nickel, and cobalt for re-use in new batteries. They are also useful for fabricating stainless steel.
- Apart from the listed objects, there is an endless list of other objects. But, overall, there's kind of hack to recycling any item or component. And no, e-waste recycling is not a one size fits all approach. However, there is a general way to go about it.

Steps for e-waste recycling

➢Recycling electronics is an often challenging activity. This is because e-scraps are typically sophisticated and manufactured from diverse elements such as metals, plastics, and glass. While this process often varies, there is a general process.

➢**Step 1: Collecting and Transporting**

➢This is the first stage of recycling e-waste. Here, recyclers place take-back booths or collection bins in specific places. When these bins get filled, the recyclers then transport the e-wastes to recycling facilities and plants.

➢**Step 2: Shredding and Sorting**

➢After collecting and transporting, the next step is to shred and sort the e-waste. The success of subsequent separation relies on shredding. And this is why efficiency is essential at this stage. Shredding involves breaking e-waste into smaller pieces for proper sorting. With the use of hands, these tiny prices get sorted and then manually dismantled. This is typically labor-intensive as waste items are, at this stage, separated to retrieve different parts.

Steps for e-waste recycling

➢After this, the materials get categorized into core materials and components. Then, these items get sorted into various categories. Typically, these category includes items that you can re-use as they are and those that require further recycling processes.

➢In any case, e-wastes are often manually sorted, while compounds such as fluorescent light, batteries, UPS batteries, and toner cartridges should not be crushed or shredded by hand.

➢**Step 3: Dust Extraction**

➢The tiny waste particles get smoothly spread via a shaking process on the conveyor belt. The smoothly spread e-waste pieces then get broken down even further. At this point, the dust gets extracted and discarded in an environmentally compliant manner. This way, there is no environmental degradation.

Steps for e-waste recycling

➢**Step 4: Magnetic Separation**

➢After this, a strong overhead magnet helps you separate steel and iron from other wastes. This way, you have successfully recycled the steel from the waste stream.

➢However, some mechanical processes may sometimes be required to separate circuit board, copper, and aluminum from other wastes particles. And this is especially where they are mostly plastic.

➢**Step 5: Water Separation**

➢After this, water separation tech becomes relevant to separate the glass from the plastic.

➢You can then send leads that contain glass to smelters to use in the production of batteries, x-ray tubes, and new CRTs.

Steps for e-waste recycling

➢**Step 6: Purification of Waste Stream**

➢The next thing is locating and extracting leftover metals from plastics to purify the waste stream further.

➢**Step 7: Preparing Recycled Materials For Sale**

➢The final stage is preparing recycled materials for sale. Here, the materials separated during the process get prepared for sale as raw materials to produce new electronics.

Benefits of e-waste recycling

1. E-waste Recycling Helps to Conserve Available Natural Resources

E-waste recycling helps recover valuable materials from electronic products that are either old or no longer used. In turn, this saves and conserves natural resources.

This is because manufacturers can now obtain raw materials from recycled waste. Consequently, there is less need to go to earth for raw materials.

So, yes, there is less need to get copper or lead or metal from mother nature. You will need to remember that these materials are not inexhaustible to realize how much good this represents.

Benefits of e-waste recycling

2. It Prioritizes Environmental Protection

As you may have observed from the processes explained above, e-waste recycling prioritizes environmental protection. It seeks to prioritize properly handling, processing and managing hazardous and toxic substances such as lead, mercury, and cadmium. All substances you may find in your e-waste stream.

As you have read, as one of the sub-steps of e-waste processing, any dust sorted from shredded particles must be disposed of in an environmentally friendly manner. This way, the hazards that these elements usually pose to our environment get substantially reduced, thanks to e-waste recycling.

Benefits of e-waste recycling

3. Creates Jobs

- E-waste recycling is creating new jobs for persons such as professional recyclers. What's more is that, by so doing, it has created a secondary market where recycled materials are the primary commodity.
- The Environmental Protection Agency released findings that show the magnitude of economic benefits that comes from e-waste recycling.
- In a year, the US's recycling activities provided 757,000 jobs, $6.7 billion tax revenues, and $36.6 billion as wages.
- By implication, for every thousand tons you recycle, there's 1.57 job created, $ 76,000 wages paid, and $ 14,101 tax revenues. It seems like a lot of benefit coming from trash, right? But there's more.

Benefits of e-waste recycling

3. Creates Jobs

- For a million laptops you recycle, you will have saved the equivalent of electric power capable of running 3657 households for one year. More, for a million cell phones, you can recover gold weighing 75 pounds, silver of 772 pounds, copper of 35,274 pounds, and palladium of 33 pounds. Amazing, right?

Benefits of e-waste recycling

4. Saves Landfills and Reduces Global Warming

- Usually, uncollected e-wastes get dumped at incinerators and landfills. By recycling e-waste, we are reducing the amount of e-wastes pilling up at these places.
- This is because two-thirds of waste on landfills are biodegradable and capable of breaking down and returning to their natural elements. As these wastes break down and decompose, they produce harmful gases (Methane and CO2) – greenhouse gases – which heavily contribute to global warming.
- landfills also pollute the water and soil in our local environment, activities like e-waste recycling, which seek to reduce these environmental concerns are not merely beneficial, but also life-saving.

Benefits of e-waste recycling

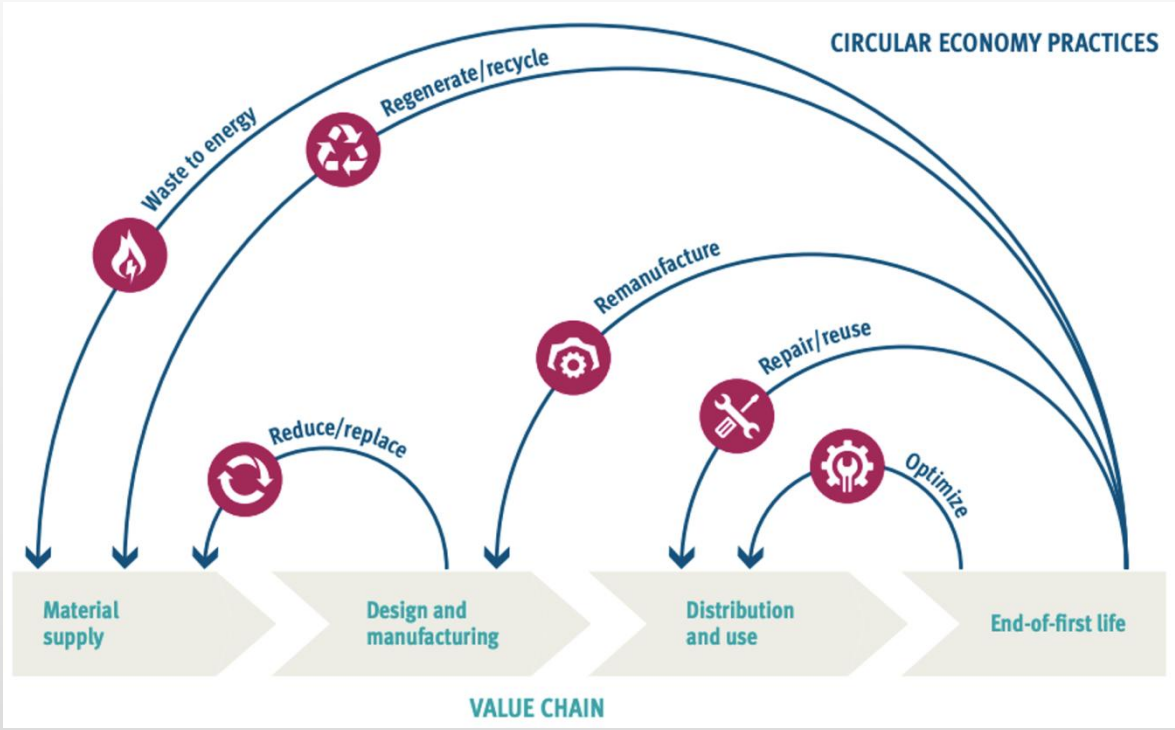
4. Saves Landfills and Reduces Global Warming

- E-waste recycling is a serious global concern for a lot of reasons. It has a strong bearing on our immediate environment as humans and life on earth generally. It even promises significant economic returns for persons, communities, and even nations.
- Many people have begun to tap from the floodgate of opportunities provided by e-waste recycling. In developing nations, entrepreneurs are taking advantage of e-recycling as a steady revenue stream and advancing environmental goals.
- Although the steps involved might be labor-intensive, they are easy to follow. From collecting and separating to preparation for sale, there are several steps involved which you can easily flow with. And of course, you too can directly benefit from e-waste recycling and also help to save our planet.

Circular economy goal

➢The goal of circular economy is to optimize circle system functions by reducing resources exiting the system. Moving toward greater circular economy can assist in meeting the resource efficiency goals set out in the Strategy for smart, inclusive, and sustainable growth.

➢For optimal sustainable development, smart cities concept is a pragmatic and balanced blend of social, economic, ecological, and other key fields.



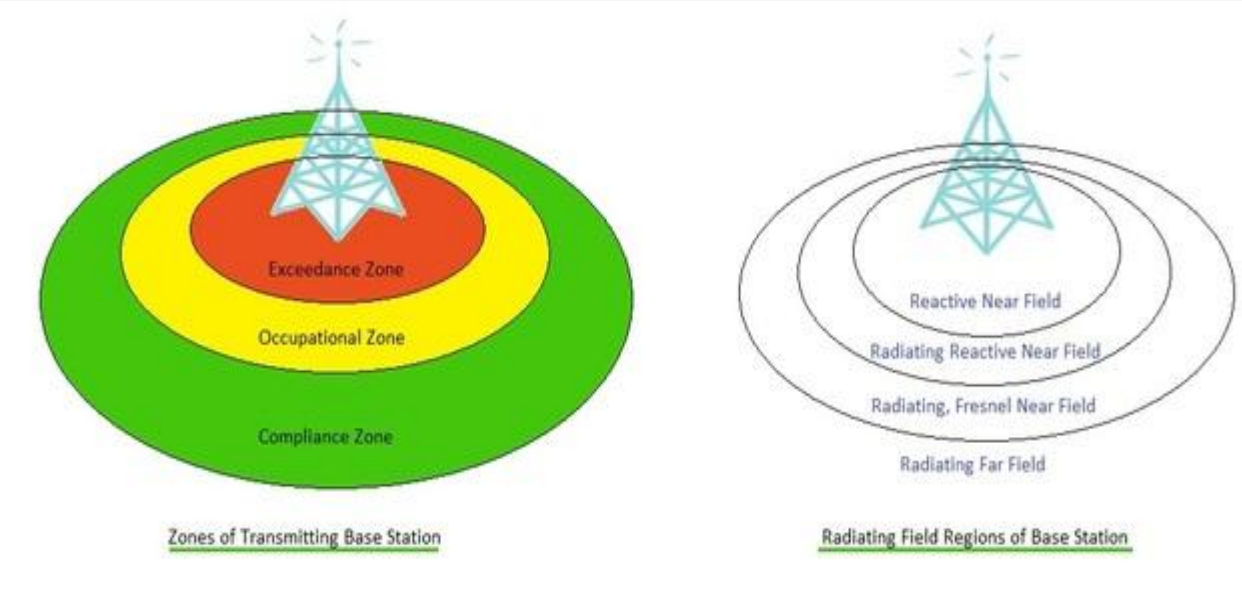
https://www.unido.org/our-focus-cross-cutting-services/circular-economy

Electromagnetic radiation sources

- Electromagnetic radiation, or EM radiation, is a fundamental form of energy that surrounds us everywhere, from the microwaves that heat our food to the radio waves that bring us music and news.
- As we know this is the age of mobile evolution across the globe. To meet the demands of increasing number of mobile and wireless devices telecom operators are deploying large number of cellular towers.
- These cellular mobile phone towers employ multiple antennas and transmit electromagnetic waves to serve the subscribers.
- Exposures to many EMFs frequencies are increasing significantly as technology advances unabated and new applications are found.
- During the past 20 years the general public has become increasingly concerned about potential adverse health effects of exposure to electric and magnetic fields.

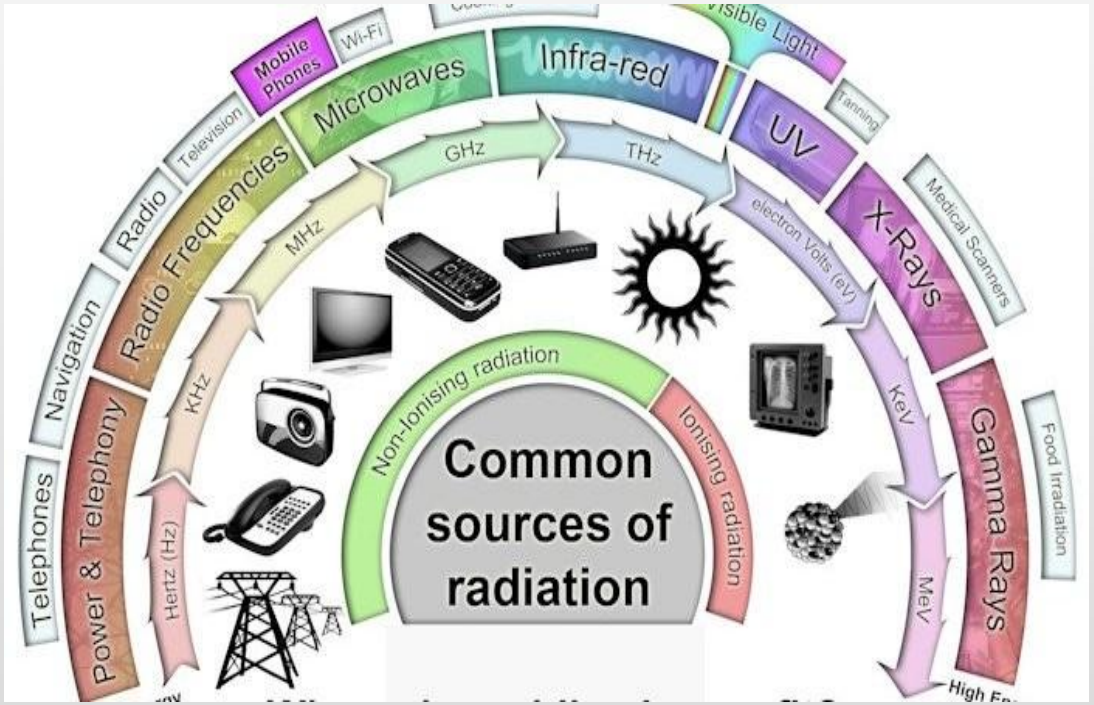
EM radiation

- The figure also depicts EM radiation field regions around the antenna.
- As mentioned there are reactive near field, radiating reactive near field, radiating fresnel near field and radiating far field regions.
- For the RF exposure assessment it is required to measure E field, H field and SAR (Specific Absorption Rate).
- All these parameters depend upon observer location (i.e. field region) and field impedance.

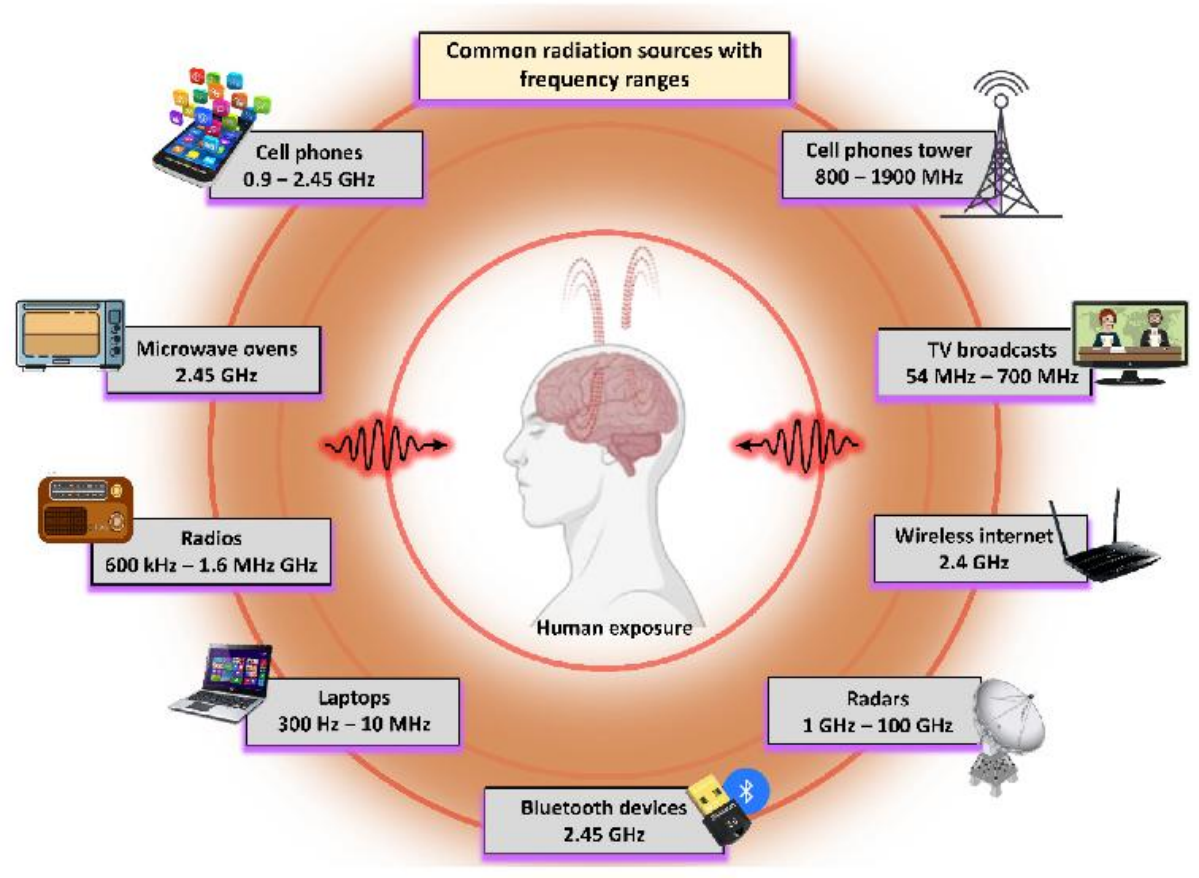


https://www.rfwireless-world.com/Articles/Electromagnetic-radiation-basics-and-EM-mitigation-techniques.html

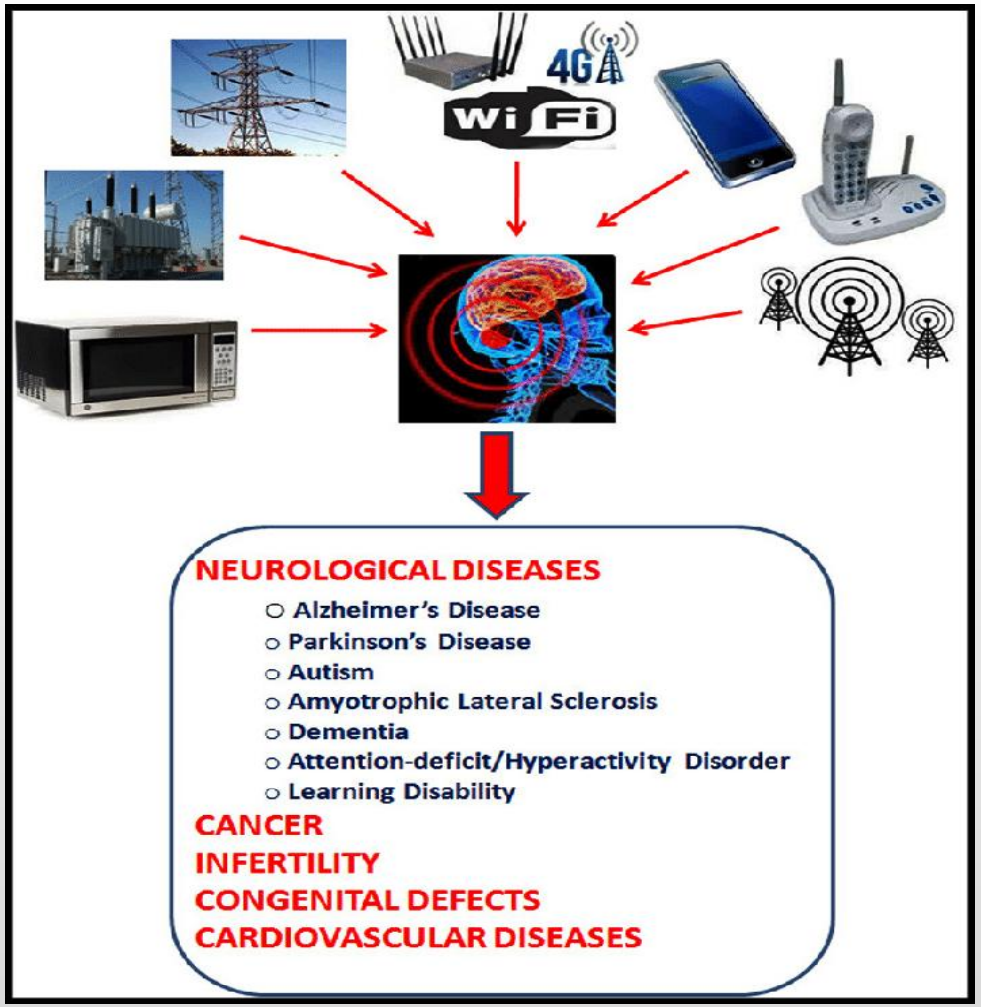
- As mentioned high level of RF exposure to humans will result into heating of the body parts (tissues) and will have adverse effects.
- This is due to the fact that humans do not have heat dissipating capabilities.
- Especially parts such as eyes and kneecaps will be affected more.
- There are two main types of radiations ionising and non ionising. We know that Human body will have large components as water.
- Ionising **EM radiation** breaks the chemical bonds of water while non-ionising EM radiation do not break the bonds. Hence non-ionising **electromagnetic radiation** is harmful for human beings.
- International Commission on non-ionising radiation protection (ICNIRP) has developed international guidelines and limits for EM radiation.



https://www.linkedin.com/pulse/mobile-phones-adverse-effect-electromagnetic-radiation-sandiso-mnguni



https://encyclopedia.pub/entry/26671



https://www.researchgate.net/publication/267743832_Effect_of_Electromagnetic_Radiations_on_Neurodegenerative_Diseases-Technological_Revolution_as_a_Curse_in_Disguise/figures?lo=1&utm_source=google&utm_medium=organic

Electromagnetic radiation mitigation technique

- There are countries where in RF exposure limit or EM radiation is monitored by government and standard agencies.
- Following are the techniques used for Electromagnetic radiation mitigation.
- Transmit Power directly depends on power density and square of E/H. Hence reduction in transmit power will result into reduction in RF radiation level. But this leads to coverage area reduction.
- As power density at a point from antenna depends on antenna height. If antenna height is increased then RF field strength at the observer point will be reduced due to increase in distance.
- Higher antenna gain and lower transmit power can be used in near proximity to the antenna.

Electromagnetic radiation mitigation technique

- Other than the mitigation techniques mentioned above, following are the basic tips for health concern to be adopted by all the mobile phone or wireless device users.
- Minimize the usage of mobile phones and/or wireless devices.
- Do not continuously wear wireless ear phone devices based on wireless technologies such as Bluetooth, Zigbee etc.
- Do not allow children to use them more. Try to use them in the safer zone areas far from the base station or cellular towers.

EMF mitigation

- Lines with current-carrying conductors positioned vertically on power line structures produce lower magnetic fields than power lines with conductors positioned horizontally.
- Fields equal to and opposite in magnitude from those emitted by the power line would be generated to cancel the fields from the power line. This approach is not very practical except for specific locations.
- The electric field component of EMF is easily shielded by most structures. However, the magnetic fields are difficult to contain with shielding. Some materials exist that have magnetic shielding characteristics, but the expense of these items is such that the application is mostly limited to small projects and specific locations.

Test system of EM pollution

- To test any place like homes, education places, streets, public places,…etc. for EM pollution, the values of magnetic field must be measured and tested against the safe values.
- This can be done by walking through the region with an ELF/EMF meter.
- If the reading is generally below 1.0 mG except near appliances, the region is wired correctly and the degree of EM pollution in such region is accepted.

Thank You

Name of the Faculty: Dr. Gulmi Chakraborty

Department of Chemistry

Mail Id: gulmichakraborty@cgu-odisha.ac.in

www.cgu-odisha.ac.in

C. V. Raman Global University
ODISHA BHUBANESWAR INDIA